

**THE APPLICATIONS OF THE CALCULUS TO
THE MATHEMATICS OF MUSICAL CONSONANCE**

A Monograph in Mathematical Acoustics

by

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Abstract

The present treatise develops a mathematically rigorous account of the relationship between the calculus and the perception of musical consonance and dissonance. The investigation proceeds from a single governing question: by what mathematical structures may the perceived ordering of musical intervals from consonant to dissonant be characterized, and to what extent does that ordering admit principled derivation from the apparatus of differential and integral calculus? The analysis is organized around four analytical instruments: the geometric sequence underlying twelve-tone equal temperament, the definite integral of the sinusoidal pressure waveform, the divergent series of integer-multiple harmonics that constitutes the spectrum of an idealized vibrating string, and the Fourier decomposition that connects time-domain waveforms to their spectral content. Each instrument is applied to a specific question in the theory of consonance, and each is situated within the broader historical and psychoacoustic literature in which mathematical music theory has been developed since antiquity.

The treatment begins with a corrected formulation of the equal-tempered frequency relation, which is shown to be a geometric sequence rather than a geometric series, and traces the consequences of that distinction for the analytic description of pitch. It then examines the integral of one positive half-cycle of the sinusoidal pressure waveform, demonstrating that the ratio of such integrals between octave-related notes recovers, but does not exceed, the ordinary inverse relationship between frequency and wavelength. The harmonic series of overtones is then introduced, and its mathematical properties—convergence of partial-harmonic energy, divergence of the underlying summation of reciprocals—are examined as they bear upon the perceived sonority of a single note. Fourier analysis is then introduced as the canonical framework within which timbre is decomposed and within which the heuristic observations of

the earlier chapters acquire their formal foundation. The investigation proceeds to a comparative analysis of alternative tuning systems, including Pythagorean, just, meantone, well-tempered, and modern equal-tempered systems, and concludes with a survey of the psychoacoustic literature on critical bandwidth, sensory dissonance, and the Plomp–Levelt and Sethares models. The investigation concludes that the simple-integer-ratio account of consonance, though historically influential and partially correct, is properly understood as a corollary of harmonic spectral structure rather than as a freestanding diagnostic principle, and that any complete account of consonance must integrate spectral, perceptual, and cultural dimensions.

Keywords: calculus, consonance, dissonance, equal temperament, Fourier analysis, harmonic series, just intonation, psychoacoustics, Pythagorean ratios, Plomp–Levelt model, Sethares dissonance curve, tuning theory.

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Chapter I

Introduction

1.1 Statement of the Problem

The pleasing or displeasing character of musical sound, characterized in the canonical literature by the binary distinction between consonance and dissonance, has occupied the attention of mathematicians, physicists, music theorists, and psychophysicists for more than two and a half millennia. From the legendary monochord experiments attributed to the Pythagorean school, through the harmonic researches of Mersenne and Sauveur, to the spectral and perceptual investigations of Helmholtz, Plomp, Levelt, and Sethares, the problem of identifying the mathematical substrate of consonance has admitted no fully satisfactory resolution. The present study examines the contribution of the calculus—the analytical apparatus of limits, derivatives, integrals, and infinite series—to the formal characterization of that substrate. Its governing thesis is that a principled and defensible account of consonance and dissonance may be developed by means of four specific instruments of the calculus: the geometric sequence underlying twelve-tone equal temperament; the definite integral of the sinusoidal pressure waveform; the divergent infinite series of integer-multiple harmonics; and the Fourier decomposition that relates time-domain signals to their spectral content. Each instrument is examined herein for its analytical reach and for its limitations.

The problem is posed in the following terms. Given any two musical tones, each represented in the idealized case as a pure sinusoidal pressure waveform of specified frequency, by what mathematical procedure may their simultaneous sounding be classified along a continuum from maximally consonant to maximally dissonant? The literature provides three

principal candidate answers, each of which the present inquiry will subject to rigorous analysis. The first is the simple-integer-ratio account, which holds that pairs of tones whose fundamental frequencies stand in ratios involving small integers are perceived as more consonant than pairs whose frequencies stand in ratios involving larger integers. The second is the spectral-coincidence account, which holds that consonance arises from the alignment of upper partials in the harmonic series generated by each fundamental. The third is the psychoacoustic account, which holds that consonance is a perceptual phenomenon best characterized by reference to the critical bandwidth of the basilar membrane and to the empirical roughness curves derived from controlled psychoacoustic experiments.

1.2 Methodology and Scope

The present treatise adopts an analytical methodology, in which each candidate account is examined on its own terms and then contrasted with the others. The principal sources of evidence are the established results of mathematical analysis—the convergence and divergence properties of relevant series, the closed-form values of definite integrals over standard intervals, and the elementary algebra of geometric sequences—together with empirical findings from the psychoacoustic literature where those findings bear upon the mathematical claims under examination. The treatise does not undertake new psychoacoustic experimentation. It restricts its empirical claims to those that have been established in the peer-reviewed literature and that admit mathematical formulation.

The scope of the investigation is restricted in three principal respects. First, the analysis is concerned with the relationship between pairs of tones rather than with the full complexity of multi-note harmony or polyphonic counterpoint. This restriction is methodological rather than substantive: the pairwise case provides the simplest setting in which the relevant mathematical

structures may be examined, and the results obtained in that setting generalize, with appropriate care, to the multi-tone case. Second, the analysis presumes that the listener is acculturated to the Western tonal tradition, in which the chromatic semitone system and the consonance–dissonance distinction are well-defined. Cross-cultural variations in the perception of consonance are addressed in Chapter VIII, but they do not constitute the principal subject of the investigation. Third, the analysis treats the musical tone in the first instance as a pure sinusoid, deferring to Chapter VI the spectral complexity introduced by real musical instruments and to Chapter VIII the perceptual complexity introduced by the human auditory system.

1.3 Governing Standards and Conventions

The mathematical conventions adopted in this work follow standard usage in mathematical analysis and acoustics. Frequencies are stated in hertz, denoting cycles per second, with the abbreviation

Hz. When a particular musical note is named, the convention is to give the note letter followed by the frequency in hertz: thus *A* 440 denotes the note *A* of frequency 440 hertz, identified in scientific pitch notation as *A*4. Where octave designations are pertinent to the analysis, the scientific pitch notation is used: thus *A*4 = 440 Hz and *A*5 = 880 Hz. The standard tuning frequency adopted throughout is that codified in International Organization for Standardization (ISO) Standard 16:1975, *Acoustics—Standard tuning frequency (Standard musical pitch)*, which establishes 440 Hz as the frequency for the note *A* in the treble stave (ISO, 1975/2017). That standard remains current as of the present writing, having been reaffirmed in 2017 and not yet superseded.

Citations and references conform to the seventh edition of the Publication Manual of the American Psychological Association (APA, 2020). Mathematical notation conforms to the

conventions of standard analysis: $\mathbb{R}$ denotes the real numbers, $\mathbb{N}$ the natural numbers, and the integral sign with explicit upper and lower bounds denotes the Riemann integral. Definite integrals are expressed in standard form, and infinite series are denoted with the conventional summation notation indexed from the stated lower bound to infinity. The derivative is denoted by Leibniz notation where the function and variable of differentiation are both salient, and by Lagrange notation where only the order of differentiation is at issue. Logarithms without an explicit base denote the natural logarithm; logarithms to base 2, which arise frequently in the theory of musical intervals, are denoted with an explicit subscript.

1.4 Organization of the Present Work

The investigation is organized into ten chapters of analysis, together with the present introductory chapter, a list of references, and three appendices providing detailed derivations, supplementary tables, and a glossary of technical terms. Chapter II provides a historical survey of the mathematical literature on consonance, from the Pythagorean tradition through the seventeenth-century work of Mersenne and Sauveur, the spectral analysis of the nineteenth century, and the psychoacoustic researches of the twentieth century. Chapter III develops the mathematical architecture of twelve-tone equal temperament, deriving the equal-tempered semitone ratio from the closure constraint of the octave and establishing the formal apparatus by means of which equal-tempered frequencies are computed. Particular attention is given to the distinction between geometric sequences and geometric series, a distinction that is occasionally elided in the introductory literature and that is corrected and clarified in the present treatment.

Chapter IV examines the sinusoidal representation of musical tones and develops the integral of one positive half-cycle as a diagnostic for octave-related notes. It demonstrates that the ratio of half-cycle integrals between octave-related notes recovers the inverse-wavelength

relationship, and it situates this observation within the broader analytical context. Chapter V introduces the harmonic series of integer-multiple overtones and examines its mathematical properties, including the divergence of the underlying summation of reciprocals and the convergence of the partial-harmonic energy. Chapter VI presents Fourier analysis as the canonical framework within which the spectral content of arbitrary periodic signals is decomposed, and shows how the observations of Chapters IV and V acquire their formal foundation within that framework. Chapter VII compares the principal historical tuning systems—Pythagorean, just, meantone, well-tempered, and equal-tempered—and analyzes the mathematical trade-offs that distinguish them. Chapter VIII surveys the psychoacoustic literature on critical bandwidth, sensory dissonance, and the Plomp–Levelt and Sethares models of perceived consonance. Chapter IX synthesizes the analytical findings of the preceding chapters and offers a critical assessment of the simple-integer-ratio account in light of the spectral and perceptual considerations developed throughout the foregoing chapters. Chapter X concludes the investigation and identifies directions for further research.

1.5 Research Questions

Five research questions structure the investigation. They are presented here in the order in which they are addressed and are returned to in the concluding chapter.

First: *to what extent does the simple-integer-ratio account of consonance, attributed to the Pythagorean tradition and codified in the tabulation of just-intonation interval ratios, admit principled derivation from the elementary apparatus of the calculus?* This question is addressed in Chapters III and IV, with critical reassessment in Chapter IX.

Second: *what is the mathematical relationship between the integral of the sinusoidal pressure waveform over one positive half-cycle and the perceived consonance of the*

corresponding musical tone? This question is addressed in Chapter IV and is shown to admit a precise but restricted answer.

Third: *in what sense does the divergent mathematical series of reciprocals correspond to the harmonic structure of an idealized vibrating string, and how does that correspondence bear upon the perception of consonance and dissonance?* This question is addressed in Chapter V and is clarified by the distinction, sometimes elided in the secondary literature, between the harmonic series in mathematics and the harmonic series in music.

Fourth: *how does Fourier analysis unify the heuristic observations of Chapters IV and V, and what does that unification reveal about the limits of the simple-integer-ratio account?* This question is addressed in Chapter VI.

Fifth: *to what extent does the simple-integer-ratio account remain defensible when measured against the psychoacoustic literature on critical bandwidth and sensory dissonance, and what mathematical refinements are required to bring it into conformity with the empirical findings of that literature?* This question is addressed in Chapters VIII and IX.

1.6 Contributions of the Present Work

The contributions of the present thesis are five in number. First, it provides a corrected formulation of the equal-tempered frequency relation, replacing the geometric-series formulation that occasionally appears in the introductory literature with the correct geometric-sequence formulation, and traces the analytical consequences of that correction. Second, it offers a rigorous evaluation of the integral-as-diagnostic argument advanced in the present author's undergraduate work on this subject, identifying both the genuine insight contained in that argument and the analytical limitations that the original treatment did not articulate. Third, it presents a unified treatment of the mathematical and musical harmonic series, clarifying a

terminological collision that has caused recurrent confusion in the pedagogical literature. Fourth, it situates the elementary integral and series analysis of Chapters IV and V within the broader framework of Fourier analysis, and shows how that framework recovers, generalizes, and refines the elementary observations. Fifth, it integrates the mathematical analysis with the psychoacoustic literature, and concludes that the simple-integer-ratio account is properly understood as a corollary of harmonic spectral structure rather than as a freestanding diagnostic principle. These contributions are individually modest. Taken together, they constitute a clarifying restatement of the mathematical basis of consonance that the author believes to be of pedagogical and scholarly utility.

Chapter II

Historical Foundations of Mathematical Music Theory

2.1 The Pythagorean Tradition

The earliest extant systematic account of the mathematical basis of consonance is attributed to the Pythagorean school of pre-Socratic Greek philosophy. The attribution to Pythagoras himself, while customary, is largely legendary; the founder of the school left no extant writings, and the doctrines now styled Pythagorean were developed and recorded over several centuries by his successors, principally Philolaus, Archytas, and the later Neoplatonic commentators. The doctrine of principal interest to the present investigation is the discovery, by whatever member or members of the school, that the musical intervals perceived by Greek listeners as consonant—the octave, the perfect fifth, and the perfect fourth—correspond to ratios of small integers when expressed as ratios of string lengths or, equivalently and inversely, as ratios of vibrational frequencies (Barker, 1989; West, 1992).

Specifically, the Pythagorean tradition held that the octave corresponds to the ratio 2:1, the perfect fifth to 3:2, and the perfect fourth to 4:3, with these three intervals being subsequently characterized as the perfect or *symphonic* consonances. The Pythagorean tetractys, a triangular arrangement of ten points in four rows of one, two, three, and four, was held by the school to contain within it the ratios of the principal consonances: 2:1, 3:2, and 4:3, together with the unison 1:1. The mystical and numerological significance attached to this arrangement by the school is not of present analytical interest. What is of present analytical interest is that the school appears to have understood, by whatever means, that the principal musical consonances

correspond to ratios of the first four positive integers, and that this correspondence admits expression by means that are amenable to systematic mathematical treatment.

The Pythagorean tradition extended this insight by means of the operation now called the cycle of fifths. Beginning with a fundamental tone and repeatedly ascending by the interval 3:2, the school generated a sequence of twelve tones that approximates, but does not exactly reproduce, the equal-tempered chromatic scale of modern Western music. The closure failure of this construction—the small but musically significant discrepancy between twelve perfect fifths and seven octaves, known as the Pythagorean comma—was not satisfactorily resolved within the Pythagorean framework and constitutes one of the central problems addressed by the alternative tuning systems examined in Chapter VII. The Pythagorean comma is approximately 23.46 cents, where one cent is one one-hundredth of an equal-tempered semitone. Its mathematical value is given by the expression $(3/2)^{12}$ divided by 2^7 , which yields 531441/524288, or approximately 1.01364.

2.2 The Aristotelian and Ptolemaic Refinements

The Pythagorean doctrine was refined and extended by subsequent classical authors, of whom Aristoxenus of Tarentum and Ptolemy of Alexandria are of principal interest to the present analysis. Aristoxenus, a student of Aristotle, departed from the Pythagorean tradition by holding that the proper standard for the description of musical intervals is the perception of the trained musical ear rather than the mathematical ratio of string lengths. His work, the *Harmonic Elements*, is the earliest extant systematic treatise on Greek music theory and provides the empirical counterweight to the more abstractly mathematical Pythagorean doctrine. Ptolemy, writing several centuries later, undertook to reconcile the two traditions by means of a detailed mathematical analysis of the various tunings then in use, including a refined system of just

intonation in which the major third is given the ratio 5:4 rather than the Pythagorean 81:64 (Barker, 1989).

The Ptolemaic refinement introduces the small prime number 5 into the system of consonance ratios, a development of considerable subsequent importance. The Pythagorean tradition had operated exclusively within the prime-limit-3 system, in which all interval ratios may be expressed as products of powers of 2 and 3. The Ptolemaic system extends this to a prime-limit-5 system, in which the prime 5 is admitted, and the major third 5:4 and the major sixth 5:3 are reclassified as consonances. This extension has the analytical consequence that the just-intonation chromatic scale is generated by combinations of the three primes 2, 3, and 5, and that the cents-values of the resulting intervals may be computed by the formula $1200 \log_2(p/q)$, where p and q are the numerator and denominator of the interval ratio. The just-intonation system, while acoustically purer than equal temperament, suffers from the same closure failure that afflicts the Pythagorean system and is unable to provide a chromatic scale that is uniformly accurate in all keys.

2.3 Medieval and Renaissance Developments

The transmission of classical Greek music theory to the Latin West proceeded principally through the encyclopedic *De Institutione Musica* of Boethius, written in the early sixth century of the common era. Boethius preserved the Pythagorean ratio system and the Ptolemaic refinements within a single integrated treatise that became the standard textbook on music theory in the medieval universities. The mathematical content of Boethius—the integer-ratio system, the cycle of fifths, the closure failure—was thereby transmitted essentially unchanged from late antiquity into the Renaissance, when it became the object of renewed analytical attention as a

consequence of the rise of polyphonic vocal music and the corresponding need for tuning systems that would accommodate modulation between keys (Palisca, 1985).

The principal Renaissance contributors to mathematical music theory were Gioseffo Zarlino, whose *Le istituzioni harmoniche* (1558) provided the first systematic Renaissance account of just intonation; Vincenzo Galilei, whose *Dialogo della musica antica et della moderna* (1581) advanced an empirically grounded critique of the Pythagorean tradition; and Marin Mersenne, whose *Harmonie universelle* (1636) provided the first systematic empirical determination of the relationship between string length, tension, and vibrational frequency. Mersenne's laws, as they are now called, state that the frequency of a vibrating string is directly proportional to the square root of its tension, inversely proportional to its length, and inversely proportional to the square root of its linear mass density. These laws constitute the first quantitative physical theory of musical sound and provide the empirical foundation for the subsequent mathematical development.

2.4 The Seventeenth and Eighteenth Centuries: Sauveur and the Birth of Acoustics

The discipline now called acoustics was named and substantially founded by Joseph Sauveur in a series of memoirs presented to the French Académie Royale des Sciences between 1700 and 1713. Sauveur was the first to recognize, name, and systematically study the phenomenon of overtones—the integer-multiple frequencies that accompany any vibrating string or column of air—and to establish that the timbre of a musical instrument is determined by the relative amplitudes of its constituent overtones. Sauveur's identification of the harmonic series of overtones, antedating Fourier's analytical theorem by more than a century, provides the empirical foundation for the spectral theory of consonance that is developed in Chapters V and VI of the present work (Cohen, 1981).

Sauveur's younger contemporary, Brook Taylor, contributed the first mathematical analysis of the vibrating string in his *Methodus Incrementorum Directa et Inversa* of 1715. Taylor derived the fundamental frequency of a string under the assumption of small transverse displacements and showed that this frequency depends on the string's tension, length, and density in precisely the manner empirically established by Mersenne. The Bernoulli family—Daniel Bernoulli and his uncle Jakob—extended Taylor's analysis in the early eighteenth century by recognizing that the general solution of the wave equation for a vibrating string admits an infinite superposition of sinusoidal modes, each corresponding to one of Sauveur's overtones. This is the substantive insight that, after subsequent generalization by Euler and Lagrange and rigorous formulation by Fourier, became the modern theory of Fourier series and the spectral analysis of musical signals.

2.5 The Nineteenth Century: Fourier, Ohm, and Helmholtz

Joseph Fourier, in his *Théorie analytique de la chaleur* (1822), established that any reasonably well-behaved periodic function may be represented as an infinite trigonometric series whose terms are sinusoids of frequencies that are integer multiples of the fundamental frequency. Fourier's theorem, although originally developed in the context of the analytical theory of heat, has acquired its principal subsequent applications in the analysis of vibrating systems and in particular in the spectral analysis of musical signals. The technical formulation of Fourier's theorem and its application to the consonance problem are developed in detail in Chapter VI.

Georg Simon Ohm, in 1843, extended Fourier's analytical theorem to the theory of auditory perception by proposing what is now called Ohm's acoustic law: the human ear performs a Fourier-like decomposition of incident sound waves into their constituent sinusoidal components, and the perceived pitch of a musical tone is determined by the fundamental

frequency of that decomposition. Hermann von Helmholtz, in his epoch-making treatise *Die Lehre von den Tonempfindungen als physiologische Grundlage für die Theorie der Musik* (1863), synthesized the spectral analysis of Fourier and Ohm with detailed empirical investigation of the resonance properties of the inner ear, and proposed the first comprehensive theory of consonance grounded in the spectral structure of musical tones. Helmholtz's theory, which holds that consonance arises from the absence of beats between the upper partials of simultaneously sounded tones, is examined in detail in Chapter VIII (Helmholtz, 1863/1954).

2.6 The Twentieth Century: Psychoacoustics and the Critical Bandwidth

The twentieth-century literature on consonance is dominated by the development of psychoacoustics as an independent experimental discipline. The principal results of this literature, for present purposes, are the identification of the critical bandwidth of the basilar membrane by Fletcher (1940) and Zwicker (1961); the empirical roughness curves of Plomp and Levelt (1965), which establish that two sinusoidal tones sound rough or dissonant when their frequency difference falls within roughly a quarter of the critical bandwidth; and the generalization of these results to arbitrary timbres by Sethares (1993, 1998), who showed that the consonance ordering of intervals depends not only on their frequency ratios but also on the spectral content of the tones being sounded.

These twentieth-century developments have the analytical consequence that the simple-integer-ratio account of consonance, while substantially correct for harmonic timbres, is properly understood as a special case of a more general perceptual phenomenon. The simple-integer-ratio ordering arises because, for harmonic timbres, simple-integer-ratio intervals produce maximum coincidence of upper partials and therefore minimum sensory roughness as measured by the Plomp–Levelt model. For inharmonic timbres—those whose overtones are not integer multiples

of a fundamental—the simple-integer-ratio account fails, and the corresponding scales of maximum consonance depart, often substantially, from the equal-tempered or just-intonation systems familiar from Western music. The mathematical implications of this generalization are developed in Chapter VIII and discussed in the synthesis presented in Chapter IX.

2.7 Summary

The historical literature surveyed in this chapter establishes three principal points that are presupposed in the analytical chapters that follow. First, the integer-ratio account of musical consonance has a continuous historical pedigree extending from antiquity to the present day, and constitutes the orthodox doctrine within the Western theoretical tradition. Second, the integer-ratio account acquired its modern mathematical foundation in the seventeenth and eighteenth centuries through the work of Mersenne, Sauveur, and the Bernoullis, and acquired its modern spectral foundation in the nineteenth century through the work of Fourier, Ohm, and Helmholtz. Third, the integer-ratio account requires significant qualification in light of twentieth-century psychoacoustic research, and is properly understood as a corollary of harmonic spectral structure rather than as a freestanding diagnostic principle. These three points constitute the historical premises of the analytical investigation that follows.

Chapter III

The Mathematical Architecture of Equal Temperament

3.1 The Closure Constraint of the Octave

The construction of twelve-tone equal temperament begins with two empirical observations and one mathematical constraint. The first empirical observation is that any musical tone of frequency f , when sounded together with a tone of frequency $2f$, is perceived as a doubling of the first tone in a sense that listeners describe as a repetition at a higher pitch. The two tones share what musicians term octave equivalence, and they are designated by the same note letter in standard musical nomenclature. The second empirical observation is that the perceptual distance between any pair of tones is, to a first approximation, a function of the logarithm of their frequency ratio rather than of their frequency difference. The mathematical constraint is that the chromatic scale, as historically developed in Western music, contains twelve distinct semitones within each octave, and that these semitones are required for the closure of the equal-tempered system to be uniform in all keys.

Given these three inputs, the equal-tempered semitone ratio r is determined uniquely by the requirement that twelve successive applications of the ratio r return a tone of exactly double the fundamental frequency. Formally:

$$r^{12} = 2, \quad \text{whence} \quad r = 2^{1/12} \approx 1.05946309\dots$$

The value 1.05946309... is the equal-tempered semitone ratio. It is the irrational number whose twelfth power is exactly 2 and which, by construction, generates a chromatic scale that closes exactly at the octave. The ratio is dimensionless: it is the ratio of two frequencies, each measured in hertz, and the units therefore cancel. The convention occasionally encountered of writing this value as "1.05946 Hz" is an elementary error and is to be avoided.

3.2 The Equal-Tempered Frequency Relation as a Geometric Sequence

The frequency of the n -th chromatic note above a fundamental of frequency f_0 is given by the relation:

$$f_n = f_0 \cdot r^n, \quad r = 2^{1/12}, \quad n \in \mathbb{Z}$$

This relation defines a geometric sequence with common ratio r and initial term f_0 . Each term of the sequence is the product of the preceding term and the constant r . The relation is to be carefully distinguished from the geometric series, which is the partial sum of the terms of a geometric sequence and is defined by the relation:

$$S_n = f_0 \cdot \frac{1 - r^{n+1}}{1 - r}, \quad r \neq 1$$

The geometric series is a sum of frequencies; the geometric sequence is an indexed family of frequencies. The two objects are mathematically distinct, and the distinction is not merely formal. The geometric series with common ratio $r = 2^{1/12} > 1$ diverges as n approaches infinity: the partial sums grow without bound, and no finite limit exists. The

geometric sequence with the same common ratio is well-defined for every integer n , including negative integers, and provides the desired frequency for every note in the chromatic scale.

It is occasionally observed in introductory treatments of this subject that the equal-tempered frequency relation may be written as an infinite series summed from $n = 0$ to infinity. This formulation is mathematically incorrect when the index n is intended to identify particular chromatic notes. The frequency of the n -th note is not the partial sum S_n but the n -th term of the sequence; the partial sum, if computed, has no clear musical interpretation and diverges in any case. The correct formulation, adopted throughout this work, is the geometric-sequence formulation given above.

3.3 Logarithmic Representation and the Cent

Because the equal-tempered semitone ratio r is irrational and the resulting frequencies are likewise irrational, the comparison of intervals is more conveniently performed in logarithmic coordinates. The cent, introduced by Alexander Ellis in the appendix to his English translation of Helmholtz's *Tonempfindungen*, is defined as one one-hundredth of an equal-tempered semitone. The cents-value of an interval whose frequency ratio is p/q is given by the relation:

$$I_{\text{cents}} = 1200 \cdot \log_2 (p/q)$$

In this representation, the octave is exactly 1200 cents, the equal-tempered semitone is exactly 100 cents, and the cents-value of an interval is additive under composition: the cents-value of the composition of two intervals is the sum of their individual cents-values. This additivity, which corresponds in frequency-ratio terms to multiplicative composition, is a direct

consequence of the logarithm function and provides one of the principal computational advantages of the cents representation. The cent is the standard unit of measurement for small interval differences in the music-theoretic literature and is used throughout the comparative analysis of Chapter VII.

3.4 Construction of the Chromatic Scale

Application of the equal-tempered frequency relation to the standard tuning fundamental of A4 = 440 Hz generates the chromatic scale tabulated in Table 1. The table provides the note name in standard musical notation, the value of n in the geometric sequence indexed from A4 (with A4 corresponding to $n = 0$), the explicit form of the frequency expression, and the resulting frequency in hertz to three decimal places. The reader is referred to Appendix B for an extended version of this table including frequencies for the octaves above and below A4.

Table 1

Frequencies of the Twelve Chromatic Notes in Equal Temperament (A4 = 440 Hz)

Note (and value of n)	n	Frequency expression	Frequency (Hz)
A4	0	$440 \cdot r^0$	440.000
A#4 / Bb4	1	$440 \cdot r^1$	466.164
B4	2	$440 \cdot r^2$	493.883
C5	3	$440 \cdot r^3$	523.251
C#5 / Db5	4	$440 \cdot r^4$	554.365
D5	5	$440 \cdot r^5$	587.330
D#5 / Eb5	6	$440 \cdot r^6$	622.254
E5	7	$440 \cdot r^7$	659.255
F5	8	$440 \cdot r^8$	698.456

F#5 / Gb5	9	$440 \cdot r^9$	739.989
G5	10	$440 \cdot r^{10}$	783.991
G#5 / Ab5	11	$440 \cdot r^{11}$	830.609
A5	12	$440 \cdot r^{12}$	880.000

Note. Frequencies are rounded to three decimal places. The value of r is $2^{(1/12)}$, and the entries are computed by direct exponentiation. The closure of the scale is exact at $n = 12$, where $r^{12} = 2$ by construction.

3.5 Properties of the Equal-Tempered Scale

The equal-tempered scale possesses three properties of analytical interest. First, it is uniform: the ratio of any two adjacent semitones is the same, namely $r = 2^{(1/12)}$, and the resulting interval structure is therefore the same in every key. This uniformity is the principal practical motivation for the adoption of equal temperament as the standard tuning system of Western music, since it permits unrestricted modulation between keys without retuning. Second, the scale is closed: twelve semitones generate exactly one octave, and the cycle of semitones therefore returns to its starting note after exactly twelve applications. Third, the scale departs from just intonation by small but musically significant amounts, with the principal departures concentrated in the major third (which is 14 cents sharp in equal temperament relative to the just 5:4) and the minor third (which is 16 cents flat). These departures are tabulated and analyzed in Chapter VII.

A fourth property, less frequently noted but of theoretical importance, is that the equal-tempered scale provides an approximation, rather than an exact realization, of every just-intonation interval other than the octave itself. This property is a consequence of the irrationality of r : no finite power of r equals a rational number other than 1 or 2, and the simple-integer-ratio intervals of just intonation are therefore approximated rather than realized by the equal-tempered

system. The implications of this approximation for the perception of consonance are examined in Chapters VII and VIII.

3.6 The Logarithmic Pitch Function

It is occasionally useful, particularly for analytical purposes, to define a continuous pitch function $p(f)$ that maps frequency to pitch in a way that is consistent with the equal-tempered scale. The standard choice is:

$$p(f) = 69 + 12 \cdot \log_2 (f/440)$$

where 69 is the MIDI note number of A4 and the constant 12 converts octaves to semitones. This function is continuous and monotonic on the positive real numbers, and it has the property that integer values of p correspond to the equal-tempered chromatic notes. Its derivative with respect to frequency is:

$$\frac{dp}{df} = \frac{12}{f \ln 2}$$

which is the rate of pitch change per unit frequency change. This derivative is inversely proportional to frequency, which is the differential expression of the empirical observation that the perceptual significance of a given frequency difference depends on the frequency at which it occurs: a one-hertz difference is musically negligible at 1000 Hz but corresponds to nearly a quarter-tone at 4 Hz. The logarithmic pitch function and its derivative provide the analytical framework within which the perceptual scaling of frequency is formally characterized.

3.7 Summary

This chapter has developed the mathematical architecture of twelve-tone equal temperament. The principal results are: that the equal-tempered semitone ratio is $r = 2^{(1/12)}$, derived uniquely from the closure constraint of the octave; that the equal-tempered frequency relation is a geometric sequence rather than a geometric series, and that this distinction is consequential for the analytical treatment of the scale; that the cent provides a logarithmic unit of measurement that converts multiplicative frequency ratios into additive perceptual quantities; that the resulting chromatic scale departs from just intonation by amounts that are tabulated in Chapter VII; and that the continuous pitch function $p(f) = 69 + 12 \cdot \log_2(f / 440)$ provides the analytical framework within which the perceptual scaling of frequency is formally characterized. These results constitute the formal apparatus presupposed in the analytical chapters that follow.

Chapter IV

Sinusoidal Representation and the Integral as Diagnostic

4.1 The Sinusoidal Idealization

The simplest mathematical idealization of a musical tone is the pure sinusoid, in which the pressure of the air at a fixed point in space varies sinusoidally in time at a constant frequency. Although no real musical tone is purely sinusoidal—every real tone of definite pitch is, by the analysis developed in Chapters V and VI, a superposition of sinusoids at integer multiples of the fundamental—the sinusoidal idealization provides the analytical foundation on which all subsequent spectral analysis is constructed. The sinusoid is, in this sense, the atomic constituent of the spectral theory, and the integral and differential properties of the sinusoid are the elementary mathematical objects of the present investigation.

The general sinusoidal pressure waveform may be written, with phase set to zero and amplitude normalized to unity, as:

$$y(x) = \sin\left(\frac{2\pi x}{\lambda}\right)$$

where x denotes the spatial coordinate along the direction of propagation and λ denotes the wavelength. The wavelength is related to the frequency f and the velocity v of sound by the elementary dispersion relation $\lambda = v / f$. At standard atmospheric conditions—a temperature of approximately 20 degrees Celsius, atmospheric pressure of approximately 101.325 kilopascals, and dry air—the velocity of sound is approximately 343 meters per second, although values in

the range 343 to 344 meters per second are encountered in the literature depending on the precise temperature assumed. The present study adopts $v = 344$ meters per second as the working value for numerical computations, consistent with the IA on which the present analysis is based, while noting that small variations in this value do not affect the qualitative conclusions of the analysis.

4.2 Superposition of Octave-Related Sinusoids

Consider the simultaneous sounding of two sinusoids whose frequencies stand in the ratio 2:1—that is, octave-related notes. With the lower note set to $A_4 = 440$ Hz and the upper note correspondingly to $A_5 = 880$ Hz, the two waveforms may be written:

$$y_1(x) = \sin\left(\frac{2\pi x \cdot 440}{344}\right), \quad y_2(x) = \sin\left(\frac{2\pi x \cdot 880}{344}\right)$$

The wavelengths of these two waveforms are $\lambda_1 = 344/440 \approx 0.7818$ meters and $\lambda_2 = 344/880 \approx 0.3909$ meters respectively, with the wavelength of A_5 being exactly half that of A_4 in conformity with the inverse relationship $\lambda = v / f$. Superposition of these two waveforms produces a periodic function whose period is equal to the longer of the two wavelengths, namely λ_1 , since within one period of A_4 the waveform of A_5 completes exactly two full cycles and returns to its starting phase. The two waveforms therefore share infinitely many coincident zero-crossings, with every zero-crossing of y_1 coinciding with a zero-crossing of y_2 , and every other zero-crossing of y_2 coinciding with a zero-crossing of y_1 .

The coincidence of zero-crossings is the spatial expression of the perceptual phenomenon of octave equivalence. Each time the lower sinusoid completes a half-cycle and crosses zero, the upper sinusoid is also crossing zero, having completed an integer number of half-cycles in the

same spatial interval. The two waveforms are, in this sense, locally synchronized at every zero-crossing of the lower waveform, and they remain in this synchronized relationship indefinitely. By contrast, two sinusoids whose frequencies do not stand in a simple integer ratio share coincident zero-crossings only at intervals whose spatial extent grows rapidly with the size of the integers involved, and in the limit of irrational frequency ratios share no exact coincidences at all.

4.3 Integration of the Positive Half-Cycle

A diagnostic of considerable analytical interest is the definite integral of one positive half-cycle of each sinusoid. For the lower note A4, this integral is taken over the spatial interval from $x = 0$ to $x = \lambda_1 / 2 = 172/440 = 43/110 \approx 0.3909$ meters:

$$A_1 = \int_0^{43/110} \sin\left(\frac{2\pi x \cdot 440}{344}\right) dx$$

The standard substitution $u = 2\pi x \cdot 440 / 344 = 110\pi x / 43$ yields $du/dx = 110\pi / 43$, so that $dx = (43/110\pi) du$. When $x = 0$, $u = 0$; when $x = 43/110$, $u = \pi$. The integral therefore reduces to:

$$A_1 = \frac{43}{110\pi} \int_0^\pi \sin u \, du = \frac{43}{110\pi} \cdot 2 = \frac{43}{55\pi} \approx 0.24886$$

For the upper note A5, the corresponding integral is taken from $x = 0$ to $x = \lambda_2 / 2 = 43/220 \approx 0.1954545\dots$ meters:

$$A_2 = \int_0^{43/220} \sin\left(\frac{2\pi x \cdot 880}{344}\right) dx = \frac{43}{220\pi} \cdot 2 = \frac{43}{110\pi} \approx 0.12443$$

The ratio of these two integrals is:

$$\frac{A_1}{A_2} = \frac{43/(55\pi)}{43/(110\pi)} = \frac{110}{55} = 2$$

That is, the integral of one positive half-cycle of A4 is exactly twice the integral of one positive half-cycle of A5. The 2:1 ratio of integrals mirrors the 1:2 ratio of frequencies and, equivalently, the 2:1 ratio of wavelengths.

4.4 Analytical Interpretation of the Half-Cycle Integral

The 2:1 ratio of half-cycle integrals between octave-related notes admits a direct analytical interpretation that has not been emphasized in the antecedent literature and that the present chapter develops at some length. The half-cycle integral of the normalized sinusoid $\sin(2\pi x / \lambda)$, taken from 0 to $\lambda/2$, may be computed in closed form as:

$$\int_0^{\lambda/2} \sin(2\pi x / \lambda) dx = \frac{\lambda}{2\pi} [-\cos(2\pi x / \lambda)]_0^{\lambda/2} = \frac{\lambda}{2\pi} \cdot 2 = \frac{\lambda}{\pi}$$

That is, the half-cycle integral of a unit-amplitude sinusoid is simply λ / π . The ratio of half-cycle integrals between two octave-related notes is therefore the ratio of their wavelengths, which by the dispersion relation is the inverse of the ratio of their frequencies. The 2:1 ratio of integrals is not a freestanding diagnostic of octave consonance: it is the geometric expression of

the wavelength relation, which is itself the inverse of the frequency relation. The two facts are equivalent.

This observation has the immediate consequence that the half-cycle integral does not provide additional analytical information beyond what is contained in the frequency ratio itself. It does not, in particular, provide an independent diagnostic of consonance: a 2:1 ratio of half-cycle integrals is exactly equivalent to a 2:1 ratio of wavelengths, which is exactly equivalent to a 1:2 ratio of frequencies, which is the defining property of the octave. The half-cycle integral therefore restates the octave relation in geometric terms, rather than supplying a freestanding mathematical criterion for the perceptual phenomenon of octave consonance.

This is not to deny that the half-cycle integral is of analytical interest. It is, on the contrary, of considerable interest precisely because it makes geometrically visible what is otherwise present only as a numerical ratio: the spatial extent of one positive half-cycle of A5 is exactly half the spatial extent of one positive half-cycle of A4, and the area enclosed beneath the corresponding sinusoid is exactly half. The visualization is pedagogically valuable. The analytical claim, however, must be carefully restricted: the half-cycle integral does not explain why octave-related notes sound consonant; it merely reformulates the wavelength relation in integral form. The explanatory work, as developed in the chapters that follow, is performed by the harmonic series of overtones and by the spectral coincidence between octave-related tones, not by the integral relation as such.

4.5 On the Improper Integral

A subsidiary observation is in order regarding the improper integral of the sinusoid over the entire positive real line. It is occasionally stated, including in the IA on which the present work is

based, that this improper integral “diverges.” The statement is technically inexact and warrants correction. The improper integral:

$$\int_0^{\infty} \sin(2\pi x/\lambda) dx$$

does not converge in the sense of the Riemann integral, but it does not diverge to infinity either. The partial integrals oscillate between 0 and λ/π without limit, and the integral therefore fails to exist in the standard sense; but the partial integrals are bounded above by λ/π and below by 0, and the integral is not “divergent” in the sense in which, for example, the integral of $1/x$ from 1 to infinity is divergent. The proper statement is that the improper integral fails to converge; the term “divergent” is reserved for integrals whose partial integrals grow without bound. The distinction is not pedantic: it bears on the analytical interpretation of the sinusoidal waveform and on the proper framing of the integral relations developed in the present chapter.

4.6 Integration Over Non-Octave Intervals

It is instructive to extend the analysis of the half-cycle integral to non-octave intervals. Consider the perfect fifth: a pair of notes whose frequencies stand in the ratio 3:2. With $A_4 = 440$ Hz set as the lower note, the upper note is at the equal-tempered E5 of approximately 659.255 Hz, or, in just intonation, at exactly 660 Hz. The half-cycle integrals are:

$$A_A = \frac{344}{440\pi} \approx 0.24886, \quad A_E^{\text{just}} = \frac{344}{660\pi} \approx 0.16591$$

The ratio of these integrals is $660/440 = 3/2$ in the just case, which again recovers the frequency ratio. For the perfect fourth (4:3), the ratio is $4/3 \approx 1.333$. For the major third (5:4 in just intonation), the ratio is $5/4 = 1.25$. In every case the ratio of half-cycle integrals equals the ratio of frequencies, which is the analytically inevitable consequence of the closed-form result that the half-cycle integral is λ/π .

What the half-cycle integral analysis does reveal—when the integration is extended over multiple half-cycles of the lower note rather than restricted to a single half-cycle—is the phenomenon of coincident zero-crossings between the two waveforms. For the octave 2:1, every zero-crossing of A4 coincides with a zero-crossing of A5. For the perfect fifth 3:2, every second zero-crossing of A4 coincides with every third zero-crossing of E5. For the perfect fourth 4:3, every third zero-crossing of A4 coincides with every fourth zero-crossing of D5. For more complex ratios, the coincidences are more sparsely distributed. This pattern of coincident zero-crossings is the spatial expression of the simple-integer-ratio account of consonance, and it admits direct visualization in the plotted waveforms. The mathematical analysis that converts this geometric observation into a quantitative theory of consonance, however, requires the apparatus of Fourier analysis developed in Chapter VI rather than the elementary integration developed in the present chapter.

4.7 Summary

This chapter has developed the elementary integration of the sinusoidal pressure waveform and has examined the half-cycle integral as a candidate diagnostic of octave consonance. The principal results are: that the half-cycle integral of a unit-amplitude sinusoid is λ/π , where λ is the wavelength; that the ratio of half-cycle integrals between two notes is therefore the ratio of their wavelengths, which is the inverse of the ratio of their frequencies; that the 2:1 ratio of half-cycle

integrals between octave-related notes is the geometric expression of the wavelength relation rather than a freestanding diagnostic of consonance; that the improper integral of the sinusoid over the positive real line fails to converge but does not diverge to infinity; and that the pattern of coincident zero-crossings between two simultaneously sounded sinusoids is the spatial expression of the simple-integer-ratio account, although its quantitative formulation requires the spectral apparatus developed in Chapter VI. The integral analysis is, in this sense, suggestive rather than dispositive: it provides geometric visualization of the relations that the spectral theory will subsequently formalize.

Chapter V

The Harmonic Series and the Theory of Overtones

5.1 A Terminological Clarification

The expression “harmonic series” is used in two distinct senses in the mathematical and music-theoretic literatures, and the failure to distinguish them with adequate clarity has been a source of recurrent confusion in the pedagogical treatment of consonance. The present chapter is concerned with both senses, and it begins with their explicit distinction.

In mathematics, the harmonic series is the divergent infinite series:

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

whose partial sums grow without bound as n approaches infinity. The divergence of this series, established by Nicole Oresme in the fourteenth century, is one of the elementary results of analysis and is the prototype of a divergent series whose terms nevertheless tend to zero. The mathematical harmonic series is a single object: a sum of reciprocals of positive integers, taken to infinity.

In music, by contrast, the harmonic series is the indexed family of integer multiples of a fundamental frequency:

$$f_n = n \cdot f_0, \quad n = 1, 2, 3, \dots$$

where f_0 is the fundamental frequency. The musical harmonic series is not a sum but a sequence; its terms are the frequencies at which an idealized vibrating string or column of air may sustain standing-wave modes, and they correspond physically to the overtones that, together with the fundamental, constitute the spectrum of a real musical tone. The musical harmonic series is divergent in the sense that its terms grow without bound, but it is not a series in the mathematical sense of a partial-sum object.

The relationship between the two senses is as follows. The musical harmonic series of frequencies, with frequencies $n \cdot f_0$, has corresponding wavelengths $\lambda_n = v / (n \cdot f_0) = \lambda_0 / n$, where $\lambda_0 = v / f_0$ is the fundamental wavelength. The wavelengths therefore form the sequence $\lambda_0 \cdot (1/n)$ for $n = 1, 2, 3, \dots$, which is the term-sequence of the mathematical harmonic series. Equivalently, the half-cycle integrals of the harmonic overtones, as computed by the methods of Chapter IV, are $\lambda_0 / (n \cdot \pi)$, and the sum of these half-cycle integrals over the first N overtones is:

$$\sum_{n=1}^N \frac{\lambda_n}{\pi} = \frac{\lambda_0}{\pi} \sum_{n=1}^N \frac{1}{n} = \frac{\lambda_0}{\pi} H_N$$

where H_N is the N -th partial sum of the mathematical harmonic series. The partial sum H_N grows like the natural logarithm of N : $H_N \approx \ln N + \gamma$, where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. The mathematical harmonic series therefore appears in the analysis of the musical harmonic series as the divergent sum of the wavelengths or, equivalently, of the half-cycle integrals of the overtones.

This relationship is suggestive but should not be over-read. The fact that the wavelengths and half-cycle integrals of the musical harmonic series correspond term-for-term with the terms of the mathematical harmonic series does not, in itself, establish any perceptual or musical

consequence of the divergence of the latter. The divergence is a mathematical fact about the partial sums; the perceptual relevance, if any, lies in the spectral content of the musical tone, which is determined by the amplitudes rather than the frequencies of the overtones. The amplitudes, in real musical instruments, decrease with increasing harmonic number, typically as $1/n$ or $1/n^2$ or faster, with the precise rate of decrease determining the timbre of the instrument.

5.2 The Vibrating String

The physical origin of the musical harmonic series is the boundary-value problem for the transverse vibration of an idealized string of length L , fixed at both ends, with uniform linear mass density ρ and constant tension T . The transverse displacement $y(x, t)$ of the string at spatial coordinate x and time t satisfies the one-dimensional wave equation:

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}, \quad c^2 = T/\rho$$

subject to the boundary conditions $y(0, t) = y(L, t) = 0$ for all t . The standard separation-of-variables solution gives the general displacement as an infinite sum of normal modes:

$$y(x, t) = \sum_{n=1}^{\infty} \left[A_n \cos\left(\frac{n\pi c t}{L}\right) + B_n \sin\left(\frac{n\pi c t}{L}\right) \right] \sin\left(\frac{n\pi x}{L}\right)$$

where the coefficients A_n and B_n are determined by the initial conditions of position and velocity. The n -th term of this sum is the n -th normal mode of the string, with spatial dependence $\sin(n \pi x / L)$ and temporal frequency $f_n = n c / (2 L)$. The fundamental frequency,

corresponding to $n = 1$, is $f_1 = c / (2 L)$; the n -th overtone is at $f_n = n \cdot f_1$, which is the musical harmonic series with fundamental f_1 .

The wave-equation derivation establishes that the musical harmonic series is not an empirical curiosity but a mathematical consequence of the elementary physics of vibrating strings. The integer multiples arise from the requirement that the spatial mode $\sin(n \pi x / L)$ satisfy the boundary conditions at both endpoints of the string: the only spatial wavenumbers that admit zeros at both $x = 0$ and $x = L$ are those of the form $n \pi / L$ for positive integer n . The corresponding temporal frequencies are then integer multiples of $c / (2 L)$, which is the musical harmonic series.

The same analysis, with appropriate modifications of the boundary conditions, applies to columns of air in pipes and to other one-dimensional vibrating systems. Open–open and closed–closed pipes support the same integer-multiple harmonic series as the string; open–closed pipes support only odd-integer multiples and therefore have a distinctive timbre, characteristic of the clarinet family. Two-dimensional and three-dimensional vibrating systems, including drumheads and concert-hall acoustics, support more complex eigenvalue problems whose solutions are not, in general, integer multiples of a fundamental; the harmonic series of integer multiples is a consequence of one-dimensional vibration with appropriate boundary conditions, and it is in this sense a one-dimensional rather than a universal phenomenon.

5.3 Tabulation of the Harmonic Series of A4

Table 2 tabulates the first sixteen harmonics of $A_4 = 440$ Hz, together with the corresponding half-cycle integrals (computed by the formula λ_n / π with the wavelength taken at $v = 344$ m/s), the nearest equal-tempered note, and the deviation in cents from that equal-tempered note. The

cents-deviation is computed by the formula $1200 \cdot \log_2(f_n / f_{ET})$, where f_n is the harmonic frequency and f_{ET} is the nearest equal-tempered frequency.

Table 2

Frequencies of the First Sixteen Harmonics of A4 ($f_0 = 440$ Hz)

n	Frequency (Hz)	Half-cycle integral	Nearest ET note	Cents-deviation
1	440.000	0.24886	A4	0.00
2	880.000	0.12443	A5	0.00
3	1320.000	0.08295	E6	+1.96
4	1760.000	0.06222	A6	0.00
5	2200.000	0.04977	C#7	-13.69
6	2640.000	0.04148	E7	+1.96
7	3080.000	0.03555	G7	-31.17
8	3520.000	0.03111	A7	0.00
9	3960.000	0.02765	B7	+3.91
10	4400.000	0.02489	C#8	-13.69
11	4840.000	0.02262	D#8	-48.68
12	5280.000	0.02074	E8	+1.96
13	5720.000	0.01914	F8	+40.53
14	6160.000	0.01778	G8	-31.17
15	6600.000	0.01659	G#8	-11.73
16	7040.000	0.01555	A8	0.00

Note. Half-cycle integrals computed by λ_n / π with $v = 344$ m/s and $\lambda_n = v / (n \cdot 440)$. Cents-deviation from the nearest equal-tempered note computed as $1200 \cdot \log_2(f_n / f_{ET})$. The harmonics that coincide exactly with equal-tempered notes are the octave-related members of the series ($n = 1, 2, 4, 8, 16$); all other harmonics depart from equal temperament by amounts ranging from approximately 2 cents (for the perfect fifth, $n = 3$) to nearly 49 cents (for the eleventh harmonic).

5.4 Overtones and the Theory of Timbre

The timbre of a musical instrument is determined, in the linear regime, by the relative amplitudes of its constituent harmonics. A pure sinusoid has only the fundamental and is perceived as a flute-like tone; a sawtooth wave has all integer harmonics with amplitudes proportional to $1/n$ and is perceived as a string-like or brass-like tone; a square wave has only the odd harmonics with amplitudes proportional to $1/n$ and is perceived as a clarinet-like tone in its idealized form. Real musical instruments produce more complex spectra that, while not exactly matching any of these idealized waveforms, share their general structure: a strong fundamental together with a characteristic distribution of upper partials whose amplitudes decrease with increasing harmonic number.

The amplitude distribution of the harmonic spectrum is what distinguishes the timbre of a violin from that of an oboe, even when both instruments are sounding the same fundamental frequency. The fundamental, by itself, determines the perceived pitch; the upper partials, in their characteristic distribution, determine the perceived quality. This decomposition of pitch from timbre is one of the principal contributions of the Fourier-analytic framework developed in Chapter VI, and it provides the technical basis for the spectral theory of consonance.

A particular instrument may, depending on excitation conditions, produce harmonics whose amplitudes deviate from the idealized $1/n$ pattern. The piano provides a notable example: real piano strings, having nonzero stiffness, support modes whose frequencies are slightly above integer multiples of the fundamental, with the deviation increasing for higher harmonics. This phenomenon, called inharmonicity, is small for the lower octaves of the piano but becomes substantial for the highest. The piano tuner compensates for inharmonicity by stretching the tuning: notes in the upper register are tuned slightly sharp, and notes in the lower register slightly

flat, relative to equal temperament. The mathematical theory of inharmonicity, developed by Fletcher (1964), is examined in Appendix A.

5.5 The Harmonic Series in the Frequency Domain

The frequency-domain representation of the harmonic series is the comb spectrum: a sequence of delta functions located at the integer multiples of the fundamental frequency, with amplitudes given by the harmonic coefficients. Formally, if the time-domain waveform is:

$$s(t) = \sum_{n=1}^{\infty} a_n \sin(2\pi n f_0 t + \varphi_n)$$

then its frequency-domain representation $S(f)$ is:

$$S(f) = \sum_{n=1}^{\infty} a_n \delta(f - n f_0) e^{i\varphi_n}$$

where δ is the Dirac delta and i is the imaginary unit. The amplitudes a_n encode the timbre of the tone, the fundamental frequency f_0 encodes the pitch, and the phases φ_n encode the precise temporal structure of the waveform. The Fourier-analytic framework developed in Chapter VI provides the technical apparatus by which time-domain waveforms are decomposed into their frequency-domain representations and conversely, and by which the spectral content of arbitrary musical signals is quantitatively characterized.

5.6 The Harmonic Series and the Theory of Consonance

The harmonic series is the spectral basis of the simple-integer-ratio account of consonance.

When two tones whose fundamental frequencies stand in a simple integer ratio are sounded simultaneously, their respective harmonic spectra overlap at every common multiple of their fundamentals. For the octave 2:1, every harmonic of the upper note coincides with an even-numbered harmonic of the lower note; the two spectra are therefore maximally aligned, and no upper partial of either tone produces a frequency that is not already present in the lower tone. For the perfect fifth 3:2, every second harmonic of the upper note coincides with every third harmonic of the lower note, producing a substantial but not maximal alignment. For the perfect fourth 4:3, every third harmonic of the upper note coincides with every fourth harmonic of the lower note, producing a still substantial alignment. For more complex ratios, the alignment is progressively reduced, and for irrational ratios there is no exact coincidence between the upper partials of the two tones.

The phenomenon of spectral alignment between harmonic tones, made precise by the Fourier-analytic framework of Chapter VI and given perceptual significance by the psychoacoustic framework of Chapter VIII, provides the mathematical substrate of the simple-integer-ratio account of consonance. The reason that octave-related notes sound consonant is that they share every upper partial; the reason that fifth-related notes sound nearly as consonant is that they share every second-or-third upper partial; the reason that dissonant intervals sound dissonant is that their upper partials produce frequencies that lie close to but not exactly upon the harmonics of the other tone, generating the sensation of roughness that the psychoacoustic literature identifies as the perceptual correlate of dissonance. This account, which the present work develops in detail in the chapters that follow, is the consensus modern account of musical

consonance and is the framework within which the elementary integral analysis of Chapter IV acquires its proper analytical significance.

5.7 Summary

This chapter has introduced the harmonic series in its two senses—mathematical and musical—and has developed the relationship between them. The principal results are: that the musical harmonic series is the indexed family of integer multiples of a fundamental frequency, derived from the wave equation for the vibrating string with appropriate boundary conditions; that the mathematical harmonic series appears in the analysis of the musical harmonic series as the divergent sum of reciprocals corresponding to the wavelengths and half-cycle integrals of the overtones; that the timbre of a musical instrument is determined by the amplitude distribution of its harmonic spectrum; that the equal-tempered chromatic scale provides a good approximation to the lower harmonics but departs substantially from the higher ones; and that the simple-integer-ratio account of consonance is properly understood as the spectral-alignment property of harmonic tones whose fundamentals stand in simple integer ratios. The Fourier-analytic framework developed in the chapter that follows provides the formal apparatus within which these observations are unified and generalized.

Chapter VI

Fourier Analysis and Spectral Decomposition

6.1 The Fourier Theorem

The Fourier theorem, established in Joseph Fourier’s *Théorie analytique de la chaleur* (1822), states that any sufficiently well-behaved periodic function may be represented as an infinite trigonometric series whose terms are sinusoids of frequencies that are integer multiples of the fundamental frequency. The precise statement of “sufficiently well-behaved” has been refined progressively since Fourier’s original work; the modern formulation, established by Dirichlet, Riemann, and Lebesgue across the nineteenth and early twentieth centuries, holds that the theorem applies to any function that is bounded, has at most a finite number of maxima and minima in any finite interval, and has at most a finite number of finite-jump discontinuities in any finite interval. These Dirichlet conditions are satisfied by every musical signal of practical interest. The Fourier theorem therefore applies, in its classical form, to the analysis of musical sound (Bracewell, 2000; Loy, 2006).

Formally, let $s(t)$ be a real-valued function of period $T = 1 / f_0$, so that $s(t + T) = s(t)$ for all t . Then $s(t)$ admits the Fourier-series representation:

$$s(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(2\pi n f_0 t) + b_n \sin(2\pi n f_0 t)]$$

where the Fourier coefficients are given by the inner products:

$$a_n = \frac{2}{T} \int_0^T s(t) \cos(2\pi n f_0 t) dt$$

$$b_n = \frac{2}{T} \int_0^T s(t) \sin(2\pi n f_0 t) dt$$

The coefficient $a_0 / 2$ is the time-average of $s(t)$ over one period and is the direct-current or DC component. The coefficients a_n and b_n for $n \geq 1$ quantify the contribution of the n -th harmonic to the signal: the amplitude of the n -th harmonic is $\sqrt{a_n^2 + b_n^2}$, and its phase is $\arctan(b_n / a_n)$. The set of all Fourier coefficients $\{a_n, b_n\}$ constitutes the spectral representation of $s(t)$, and the recovery of $s(t)$ from its spectral representation is the synthesis problem; the computation of the spectral representation from $s(t)$ is the analysis problem.

6.2 The Complex-Exponential Form

A computationally and theoretically more convenient form of the Fourier series uses complex exponentials in place of sines and cosines:

$$s(t) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi i n f_0 t}$$

with complex Fourier coefficients:

$$c_n = \frac{1}{T} \int_0^T s(t) e^{-2\pi i n f_0 t} dt$$

The complex-exponential form has three principal analytical advantages. First, the index n ranges over all integers, positive and negative, allowing the representation to encompass both forward-propagating and backward-propagating waveforms in a unified notation. Second, the complex coefficients c_n encode amplitude and phase in a single complex number, with $|c_n|$ giving the amplitude and $\arg(c_n)$ giving the phase. Third, the analysis and synthesis operations are unified into a single operation on the complex exponentials, which simplifies the derivation of identities such as Parseval's theorem and the convolution theorem. The relationship between the trigonometric and complex-exponential coefficients is given by $c_n = (a_n - i b_n) / 2$ for $n \geq 1$ and $c_{-n} = (a_n + i b_n) / 2$, with $c_0 = a_0 / 2$.

6.3 The Fourier Transform

For functions that are not periodic—a class that includes the pressure waveforms produced by real musical instruments, which are approximately periodic over short intervals but evolve in time over longer intervals—the Fourier-series representation is replaced by the Fourier transform:

$$S(f) = \int_{-\infty}^{\infty} s(t) e^{-2\pi i f t} dt$$

with inverse:

$$s(t) = \int_{-\infty}^{\infty} S(f) e^{2\pi i f t} df$$

The function $S(f)$ is the frequency-domain representation of $s(t)$; it is, in general, complex-valued, with $|S(f)|$ giving the amplitude spectrum and $\arg(S(f))$ giving the phase spectrum. The Fourier transform reduces to a sum of delta functions when $s(t)$ is periodic, recovering the Fourier-series representation as a special case. The class of functions for which the Fourier transform exists in the standard sense includes all functions in the Lebesgue space $L^1(\mathbb{R})$; the extension to $L^2(\mathbb{R})$ by means of the Plancherel theorem, and to tempered distributions by means of the theory of Schwartz, accommodates the full range of functions of acoustical interest.

The convolution theorem, a principal analytical tool in the application of the Fourier transform, states that the Fourier transform of a convolution is the product of the Fourier transforms of the operands:

$$\mathcal{F}[s \cdot g] = \mathcal{F}[s] * \mathcal{F}[g], \quad \mathcal{F}[s * g] = \mathcal{F}[s] \cdot \mathcal{F}[g]$$

where the asterisk denotes convolution and $\mathcal{F}$ denotes the Fourier transform. The first form is the modulation theorem and is essential to the analysis of amplitude-modulated signals; the second form is the convolution theorem proper and is essential to the analysis of linear time-invariant filtering, which includes the room acoustics and the resonance behavior of musical instruments.

6.4 Parseval's Theorem and the Energy Spectrum

Parseval's theorem, in its Fourier-transform form, states that the total energy of a signal is preserved by the Fourier transform:

$$\int_{-\infty}^{\infty} |s(t)|^2 dt = \int_{-\infty}^{\infty} |S(f)|^2 df$$

The integrand on the left is the instantaneous power of the time-domain signal; the integrand on the right is the power spectral density of the frequency-domain signal. The energy spectrum $|S(f)|^2$ therefore provides a complete account of the energetic content of the signal, distributing the total energy across the frequency axis. For periodic signals, the analogous identity is Parseval's identity for Fourier series:

$$\frac{1}{T} \int_0^T |s(t)|^2 dt = \left(\frac{a_0}{2}\right)^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

which expresses the mean-square value of the signal as the sum of the squared amplitudes of its harmonics. The mean-square value is the perceptually relevant measure of signal intensity, and Parseval's identity therefore connects the time-domain perception of loudness to the frequency-domain decomposition of the signal into harmonic components.

6.5 The Discrete Fourier Transform and the FFT

For digital analysis of recorded musical signals, the continuous Fourier transform is replaced by the discrete Fourier transform (DFT), which operates on a finite sequence of N samples and produces a finite sequence of N frequency-domain values:

$$S[k] = \sum_{n=0}^{N-1} s[n] e^{-2\pi i k n / N}, \quad k = 0, 1, \dots, N-1$$

with inverse:

$$s[n] = \frac{1}{N} \sum_{k=0}^{N-1} S[k] e^{2\pi i k n / N}$$

Direct computation of the DFT requires N^2 complex multiplications, which is computationally prohibitive for the sample sizes encountered in audio analysis. The Cooley–Tukey fast Fourier transform (FFT) algorithm, published in 1965 but anticipated by Gauss in unpublished work of 1805, reduces the computational complexity to $N \log_2 N$ by recursive decomposition of the DFT into successively smaller DFTs. The FFT algorithm is the computational foundation of all modern digital audio analysis, and it makes practical the spectral analysis of arbitrary musical signals at sample rates and analysis windows that would otherwise be infeasible.

6.6 The Spectrogram

The Fourier transform provides a complete account of the frequency content of a signal but no account of its temporal evolution: the spectrum $S(f)$ integrates over all of time, and the temporal location of any spectral feature is lost. For musical signals, which evolve in time on multiple scales—the pitch of a sustained note, the attack and decay envelope of a struck note, the temporal succession of notes in a phrase—a time-resolved spectral analysis is required. The standard tool for this purpose is the spectrogram, which is the squared magnitude of the short-time Fourier transform:

$$S(t, f) = \int_{-\infty}^{\infty} s(\tau) w(\tau - t) e^{-2\pi i f \tau} d\tau$$

where $w(\tau - t)$ is a windowing function localized in time near $\tau = t$. The spectrogram $|S(t, f)|^2$ is a two-dimensional function of time and frequency, representing the local spectral content of the signal in a temporal neighborhood of each instant. The Heisenberg uncertainty principle for the Fourier transform imposes a fundamental trade-off between temporal and spectral resolution: the product $\Delta t \cdot \Delta f$ is bounded below by a constant that depends on the choice of window function, and the analyst must choose the window to balance temporal and spectral precision according to the analytical requirements of the application.

6.7 Fourier Analysis and the Theory of Consonance

The Fourier-analytic framework provides the formal foundation for the spectral theory of consonance introduced informally in Chapter V. The principal analytical result of this chapter, for present purposes, is that any periodic musical tone may be represented as an infinite sum of sinusoids whose frequencies are integer multiples of the fundamental, with amplitudes and phases determined by the inner products of the time-domain signal with the corresponding basis functions. The spectral content of the tone is therefore completely characterized by the sequence of Fourier coefficients $\{c_n\}$, and the simultaneous sounding of two tones is, in the linear regime, represented in the frequency domain by the superposition of their respective coefficient sequences.

Consider the simultaneous sounding of two harmonic tones with fundamentals f_A and f_B . The combined spectrum consists of two harmonic series, with frequencies $n \cdot f_A$ and $m \cdot f_B$ for positive integers n and m . The two series share a common frequency precisely when $n \cdot f_A = m \cdot f_B$, that is, when $f_A / f_B = m / n$ for some positive integers m and n . The simplest such coincidences arise when m / n is a simple ratio: $m / n = 2$ gives the octave, $m / n = 3 / 2$ gives the perfect fifth, $m / n = 4 / 3$ gives the perfect fourth, and so on. For each such ratio, the

corresponding spectral alignment is dense: the fraction of harmonics of one tone that coincide with harmonics of the other is $1 / \max(m, n)$, and the fraction of “isolated” harmonics—those that do not coincide with any harmonic of the other tone—is $1 - 1 / \max(m, n)$.

For the octave $2 / 1$, every harmonic of the upper tone coincides with a harmonic of the lower tone, and the fraction of coincident harmonics is 1. For the perfect fifth $3 / 2$, every second harmonic of the upper tone coincides with every third harmonic of the lower, and the fraction of coincident harmonics is $1/3 = 0.333$. For the perfect fourth $4 / 3$, the fraction is $1/4 = 0.25$. For the major third $5 / 4$, the fraction is $1/5 = 0.20$. For the tritone $45 / 32$, the fraction is $1/45 \approx 0.022$. The fraction of coincident harmonics provides a quantitative measure of spectral alignment, and this measure correlates with the perceived consonance of the interval in the manner predicted by the simple-integer-ratio account.

The Fourier-analytic framework therefore recovers the simple-integer-ratio account of consonance as a consequence of spectral alignment between harmonic tones. The framework also reveals the limitations of the simple-integer-ratio account, in two principal respects. First, the account presupposes that the tones being sounded have harmonic spectra—that is, that their Fourier coefficients are concentrated at integer multiples of a fundamental. For inharmonic tones, whose spectral content is not integer-multiple, the simple-integer-ratio account does not apply, and the consonance ordering depends on the specific spectral content of the tones rather than on their fundamental frequencies. Second, the account presupposes that the relevant perceptual variable is the alignment of upper partials, rather than some other property of the combined spectrum. The Plomp–Levelt model of Chapter VIII provides a more refined account, in which consonance depends on the sensory roughness produced by pairs of partials at small frequency differences, where “small” is defined relative to the critical bandwidth of the basilar membrane.

6.8 Summary

This chapter has developed the Fourier-analytic framework within which the elementary observations of Chapters IV and V acquire their formal foundation. The principal results are: that any periodic function satisfying the Dirichlet conditions admits a representation as an infinite trigonometric series whose terms are sinusoids at integer multiples of the fundamental frequency; that the Fourier transform extends this representation to non-periodic functions; that Parseval's theorem expresses the conservation of energy under the Fourier transform; that the discrete Fourier transform and its fast-algorithm implementation provide the computational foundation of modern digital audio analysis; that the spectrogram provides a time-resolved spectral representation subject to a fundamental temporal-spectral trade-off; and that the Fourier-analytic framework recovers the simple-integer-ratio account of consonance as a consequence of spectral alignment between harmonic tones, while revealing the limitations of that account when applied to inharmonic timbres or when refined by the psychoacoustic considerations developed in Chapter VIII.

Chapter VII

Alternative Tuning Systems and Their Trade-offs

7.1 The Closure Problem

The central problem of musical tuning theory is the closure problem: no tuning system constructed from simple-integer-ratio intervals can close exactly upon itself after a finite number of steps, because no finite product of small primes can equal an integer power of 2 unless the small primes are themselves limited to 2. This mathematical fact has the analytical consequence that every tuning system based on simple integer ratios must either (i) accept the closure failure and live with the resulting tuning errors, (ii) distribute the closure error across the cycle by detuning some or all of the intervals from their just-intonation values, or (iii) abandon the constraint of pure-ratio intervals altogether and adopt an equal-tempered system in which all intervals are detuned by small and uniform amounts. The historical tuning systems examined in this chapter represent the principal compromises that have been adopted to address the closure problem.

Formally, the closure problem may be stated as follows. Let $p_1, p_2, \dots, p_k$ be the prime numbers admitted as factors in the interval ratios of the tuning system, and let $n_1, n_2, \dots, n_k$ be the exponents of these primes in the cumulative interval that closes the cycle. The closure condition is:

$$p_1^{n_1} \cdot p_2^{n_2} \cdots p_k^{n_k} = 2^m$$

for some integer m . By the fundamental theorem of arithmetic, this equation has solutions if and only if the only prime admitted is 2 itself, in which case any power of 2 satisfies the condition trivially. For systems that admit the prime 3, as the Pythagorean cycle does, the closure condition $(3/2)^n = 2^m$ has no integer solutions, because the right side is a power of 2 and the left side has factors of 3 in its numerator that cannot be canceled. The closure failure is therefore a fundamental mathematical consequence of the prime structure of the integers, not an empirical limitation of any particular tuning system.

7.2 Pythagorean Tuning

Pythagorean tuning constructs the chromatic scale by successive application of the perfect fifth 3:2 and its inversion the perfect fourth 4:3, reducing each result to within an octave by multiplication or division by 2. Twelve applications of the perfect fifth generate a sequence of twelve distinct notes, which is then identified with the chromatic scale. The Pythagorean comma is the discrepancy between twelve perfect fifths and seven octaves:

$$\frac{(3/2)^{12}}{2^7} = \frac{531441}{524288} \approx 1.01364$$

which in cents is $1200 \cdot \log_2(531441/524288) \approx 23.46$ cents. The Pythagorean cycle therefore does not close: a tone reached by twelve perfect fifths is approximately a quarter-semitone sharp relative to the tone reached by seven octaves. This discrepancy is the Pythagorean comma, and it is the prototype of the closure failures that afflict all simple-integer-ratio tuning systems.

The principal analytical property of Pythagorean tuning is that its perfect fifths and perfect fourths are exact, while its thirds and sixths are mistuned. The Pythagorean major third is $(9/8)^2 = 81/64 \approx 1.2656$, which differs from the just-intonation major third $5/4 = 1.25$ by the syntonic comma $81/80 \approx 1.0125$, or approximately 21.51 cents. This mistuning of the major third is musically conspicuous and was tolerable in the medieval and early Renaissance musical idiom but became unacceptable with the rise of triadic harmony in the fifteenth and sixteenth centuries. The transition from Pythagorean to just intonation is, in this sense, the analytical reflection of a corresponding musical transition from modal counterpoint to triadic harmony.

7.3 Just Intonation

Just intonation, in its standard 5-limit form, constructs the chromatic scale from products of the primes 2, 3, and 5. The principal intervals of just intonation are:

octave 2:1; P5 3:2; P4 4:3; M3 5:4; m3 6:5

M6 5:3; m6 8:5; M2 9:8; M7 15:8; m7 9:5

These intervals are the basis of the pure consonance for which just intonation is justly celebrated. The just-intonation triads—major triad 4:5:6, minor triad 10:12:15—produce sonorities of exceptional purity, with maximum spectral alignment between the upper partials of the constituent tones. The just-intonation chromatic scale, however, suffers from a more severe closure problem than Pythagorean tuning: not only does the cycle of fifths fail to close, but the same chromatic note (say, D) has different just-intonation values depending on the harmonic

path by which it is reached. A D reached as the perfect fifth above G is at $9/8$ above C, that is, at 1.125; but a D reached as the minor third below F is at $6/5$ below $4/3$, that is, at $10/9$ above C, that is, at 1.1111. The two values differ by the syntonic comma $81/80$, and the choice between them depends on the harmonic context.

Just intonation is therefore not, properly speaking, a fixed tuning system but a family of tunings that depend on the harmonic context of the music being performed. For music that does not modulate—that is, music that remains in a single key throughout—just intonation provides a stable and pure tuning. For music that modulates to distant keys, just intonation requires retuning of one or more notes at each modulation, which is impractical for fixed-pitch instruments and which motivated the development of the alternative tuning systems examined below. For unaccompanied vocal ensembles, by contrast, just intonation may be adopted in practice without difficulty, because the singers may adjust their pitches continuously according to the harmonic context.

Table 3 reports the principal intervals of just intonation, ordered from most to least consonant by reference to the simplicity of the integer ratio. The ordering is the orthodox consonance ordering of the Western tradition; its empirical justification, by reference to spectral alignment and sensory roughness, is developed in Chapters VI and VIII.

Table 3

Pythagorean and Just-Intonation Interval Ratios Ordered from Most to Least Consonant

Interval	Ratio	Classification
Perfect octave	2 : 1	Perfect consonance
Perfect fifth	3 : 2	Perfect consonance
Perfect fourth	4 : 3	Perfect consonance

Major sixth	5 : 3	Imperfect consonance
Major third	5 : 4	Imperfect consonance
Minor third	6 : 5	Imperfect consonance
Minor sixth	8 : 5	Imperfect consonance
Minor seventh	9 : 5	Mild dissonance
Major second	9 : 8	Mild dissonance
Major seventh	15 : 8	Strong dissonance
Minor second	16 : 15	Strong dissonance
Tritone	45 : 32	Strong dissonance

Note. The classification reflects the conventional Western theoretical tradition; perfect consonances are those whose ratios involve the integers 1 through 4, imperfect consonances those involving the integer 5, and dissonances those involving larger integers. The ordering within each class is a matter of theoretical convention rather than empirical universality, and individual listeners and traditions may rank these intervals somewhat differently in detail. The tritone, with its 45:32 ratio, is the most spectrally isolated of the intervals tabulated, owing to the comparatively large integers in its just-intonation ratio.

7.4 Meantone Temperament

Meantone temperament represents the principal Renaissance compromise between just intonation and the closure requirement of fixed-pitch instruments. The system narrows the perfect fifth from its just value of $3/2$ to a slightly smaller value chosen so that four successive applications produce a just major third. In the standard quarter-comma meantone, the perfect fifth is narrowed by one-quarter of the syntonic comma $81/80$:

$$\text{fifth}_{\text{QC}} = (3/2) \cdot (80/81)^{1/4} \approx 1.4953$$

which in cents is approximately 696.58 cents, compared with the just 701.96 cents and the equal-tempered 700.00 cents. Four meantone fifths, reduced by two octaves, produce the major third $(\text{fifth_QC})^4 / 4 = 5/4$ exactly, which is the just-intonation major third. The system therefore provides pure major thirds at the cost of slightly narrowed fifths.

Meantone temperament has the analytical property that its major thirds are pure but its perfect fifths are slightly flat; the closure problem is not eliminated but is concentrated in a single grossly mistuned fifth, the so-called “wolf fifth,” which is conventionally placed in a position where it is rarely used in the music being performed. The wolf fifth in quarter-comma meantone is approximately 738.0 cents wide, which is approximately 36 cents wider than the equal-tempered fifth and is musically intolerable when sounded directly. The placement of the wolf fifth therefore constrains the keys that may be used in meantone music; the system is well-suited to keys near C and F major, where the wolf is far from the principal triads, but becomes increasingly problematic in keys with many sharps or flats.

7.5 Well Temperament

The well temperaments developed in the late seventeenth and early eighteenth centuries—those of Werckmeister, Kirnberger, Vallotti, and Young, among others—distribute the closure error of the cycle of fifths unevenly across the twelve fifths, with the principal fifths near C and G receiving smaller departures from just intonation and the more remote fifths receiving larger departures. The result is a tuning system in which all twelve keys are usable, but in which each key has its own characteristic intonation—major keys with few accidentals are purest, and keys with many accidentals are progressively more tempered. This unequal distribution of error across keys is the analytical basis for the seventeenth- and eighteenth-century convention of key affect, in which each key was held to have its own emotional character. The convention persisted into

the nineteenth century but lost its theoretical foundation with the universal adoption of equal temperament, in which all keys are intonationally identical.

The Werckmeister III temperament of 1691, perhaps the most influential of the well temperaments, distributes the Pythagorean comma across four of the twelve fifths (C–G, G–D, D–A, and B–F♯) and leaves the remaining eight fifths in their pure 3:2 form. The cents-values of the resulting interval positions admit explicit computation and are tabulated in standard references (Barbour, 1951; Jorgensen, 1991). The bearing of this discussion on the present analysis is that the well temperaments represent the principal pre-equal-tempered compromise between key uniformity and consonance purity, and that the eventual triumph of equal temperament reflects a calculated preference for uniformity over local purity—a preference that, as the analytical apparatus of the foregoing chapters makes clear, has the consequence that no key in equal temperament reproduces the spectral alignment of pure just intonation, but that all keys exhibit the same small departure from such alignment.

7.6 Equal Temperament

Twelve-tone equal temperament, the standard tuning system of modern Western music, has been examined in detail in Chapter III. Its principal analytical property, for the comparative purposes of the present chapter, is that all twelve intervals other than the octave are approximated rather than realized: the perfect fifth is 1.96 cents flat relative to just intonation, the major third is 13.69 cents sharp, the minor third is 15.64 cents flat, and so on. These departures are small but spectrally significant: the equal-tempered major third generates a beat rate of approximately 7 beats per second when sounded with the just major third in the octave above middle C, which is at the threshold of perceptual roughness. The equal-tempered minor third is somewhat worse, generating approximately 8 beats per second under analogous conditions.

The principal compensating advantage of equal temperament is that the closure problem is resolved exactly: twelve equal-tempered semitones make exactly one octave, by construction, and the cycle of fifths therefore closes without residual error. Every key in equal temperament is intonationally identical to every other, and modulation between keys is unconstrained by tuning considerations. This uniformity is the essential precondition for the music of the common-practice period and for the subsequent developments of late-Romantic chromaticism, twentieth-century atonal music, and serial composition; without it, the freedom of modulation that defines the harmonic style of Bach, Beethoven, Brahms, Wagner, and Schoenberg would not be available.

7.7 Comparative Tabulation of Tuning Systems

Table 4 tabulates the cents-deviation of the principal intervals in equal temperament from their just-intonation values, and Table 5 tabulates the closure error of the cycle of fifths in each of the principal tuning systems examined above.

Table 4

Cents-Deviation of Equal-Tempered Intervals from Just-Intonation Intervals

Interval	Just ratio	Just (cents)	ET (cents)	Δ (cents)
Unison	1/1	0.00	0	0.00
Minor 2nd	16/15	111.73	100	−11.73
Major 2nd	9/8	203.91	200	−3.91
Minor 3rd	6/5	315.64	300	−15.64
Major 3rd	5/4	386.31	400	+13.69
Perfect 4th	4/3	498.04	500	+1.96
Tritone	45/32	590.22	600	+9.78

Perfect 5th	3/2	701.96	700	−1.96
Minor 6th	8/5	813.69	800	−13.69
Major 6th	5/3	884.36	900	+15.64
Minor 7th	9/5	1017.60	1000	−17.60
Major 7th	15/8	1088.27	1100	+11.73
Octave	2/1	1200.00	1200	0.00

Note. Just-intonation cents-values computed as $1200 \cdot \log_2(p/q)$. The principal consonant intervals—octave, perfect fifth, perfect fourth—deviate from equal temperament by less than 2 cents and are perceptually nearly indistinguishable. The thirds and sixths, which are the principal sites of equal-tempered intonation error, deviate by approximately 14 to 16 cents, which is well above the perceptual threshold for trained musicians.

Table 5

Cycle-of-Fifths Closure Error Across Tuning Systems

System	Fifth (cents)	12-fifth error	Wolf?
Pythagorean	701.96	+23.46	Yes
Just intonation	701.96 (varies)	+23.46	Yes
Quarter-comma meantone	696.58	−41.06	Yes
Werckmeister III	698.05–701.96	0.00 (distributed)	No
12-TET	700.00	0.00	No

Note. The “12-fifth error” column gives the cents-difference between twelve fifths and seven octaves; a positive value indicates that the twelve fifths exceed the seven octaves, a negative value that they fall short. Pythagorean and just intonation share the same 23.46-cent Pythagorean comma. Meantone, with its narrowed fifths, has a larger absolute closure error in the opposite direction. Werckmeister and 12-TET both close exactly; the former does so by distributing the error unequally across the cycle, the latter by distributing it equally.

7.8 Microtonal Equal Temperaments

The twelve-tone equal-tempered system is one of an infinite family of equal-tempered systems indexed by the number of equal divisions of the octave. Other members of this family that have received serious analytical and compositional attention include the 19-tone equal temperament, the 31-tone equal temperament, and the 53-tone equal temperament. The 31-tone temperament, advocated by Christiaan Huygens in the seventeenth century and revived in the twentieth by Adriaan Fokker, provides remarkably accurate approximations to the just-intonation intervals: its major third is 0.78 cents flat relative to $5/4$, and its perfect fifth is 5.18 cents flat relative to $3/2$. The 53-tone temperament provides even more accurate approximations, with its perfect fifth differing from $3/2$ by only 0.07 cents, but at the cost of substantially greater complexity in the constituent intervals. The general theoretical question of which equal-tempered systems provide good approximations to the just-intonation intervals is a question in the theory of continued fractions and Diophantine approximation: the rational approximations to $\log_2(3/2) \approx 0.5849625$ with denominators 12, 41, 53, and so on, generate the equal-tempered systems whose perfect fifths most closely approximate just intonation. The 12-tone system, in this analytical light, is the smallest equal-tempered system that provides a satisfactory approximation to the principal just intervals, and its adoption as the standard system of Western music is the consequence of a Diophantine approximation problem that admits an elegant solution at the value 12.

7.9 Summary

This chapter has surveyed the principal historical tuning systems and has analyzed the closure problem that motivates the comparisons among them. The principal results are: that no simple-integer-ratio tuning system can close exactly upon itself; that Pythagorean tuning preserves the

perfect fifth at the cost of mistuned thirds; that just intonation preserves all principal intervals locally at the cost of harmonic ambiguity and impractical retuning requirements; that meantone temperament narrows the fifth to recover the pure major third at the cost of a wolf fifth; that the well temperaments distribute the closure error unevenly across the cycle to permit use of all keys with characteristic intonation; and that equal temperament resolves the closure problem exactly at the cost of small but spectrally significant departures from just intonation in every interval other than the octave. The 12-tone equal-tempered system is the smallest equal-tempered system that provides a satisfactory Diophantine approximation to the principal just intervals, and its adoption as the standard Western tuning system reflects a calculated preference for uniformity over local purity. The bearing of this comparative analysis on the theory of consonance developed in the present work is that the simple-integer-ratio account is best understood as an analytical ideal that is approximated, with varying degrees of fidelity, by each of the historical tuning systems; no system realizes the ideal exactly, and the question of which system best approximates the ideal admits no universal answer but depends on the music being performed and the aesthetic priorities of the performer.

Chapter VIII

Psychoacoustic Models of Consonance and Dissonance

8.1 The Critical Bandwidth

The mathematical and spectral analyses of the preceding chapters establish the structural foundations of the simple-integer-ratio account of consonance, but they do not, in themselves, explain why simple-integer-ratio intervals are perceived as more consonant than complex-ratio intervals. The perceptual explanation requires reference to the physiology of the human auditory system, and in particular to the frequency-resolving properties of the cochlea, where the mechanical resonance of the basilar membrane converts incident sound waves into the neural signals that are subsequently processed by the central auditory pathway. The principal psychoacoustic quantity for present purposes is the critical bandwidth, which characterizes the frequency-resolving precision of the basilar membrane at any given frequency.

The critical bandwidth was identified empirically by Fletcher (1940) and quantified systematically by Zwicker (1961) in a series of experiments in which subjects were asked to detect a sinusoidal tone embedded in band-limited noise. Fletcher found that the detection threshold for the tone depended on the bandwidth of the masking noise only when that bandwidth fell below a critical value; for bandwidths above the critical value, the threshold was independent of bandwidth and depended only on the spectral density of the noise. The critical bandwidth, in this sense, is the bandwidth of an auditory filter at the frequency in question: tones within the critical bandwidth interact perceptually, while tones separated by more than the critical bandwidth are processed in independent channels.

The critical bandwidth, denoted $CB(f)$, is a function of frequency. At low frequencies, the critical bandwidth is approximately constant at about 100 Hz; at frequencies above approximately 500 Hz, the critical bandwidth is approximately proportional to the frequency itself, with a proportionality constant of about 0.2. The standard analytical approximation, attributable to Glasberg and Moore (1990), is:

$$CB(f) = 24.7 \cdot (4.37 f / 1000 + 1) \text{ (Hz, with } f \text{ in Hz)}$$

This formula, called the equivalent rectangular bandwidth (ERB) of the auditory filter, provides an analytically tractable approximation to the empirically measured critical bandwidths. At 440 Hz, the ERB is approximately 71 Hz; at 880 Hz, approximately 119 Hz; at 1760 Hz, approximately 215 Hz. The critical bandwidth scales logarithmically with frequency in the same general manner as musical intervals scale logarithmically with frequency, which is one reason that the perceptual structure of musical intervals approximately corresponds to the structure of the chromatic scale.

8.2 The Plomp–Levelt Model of Sensory Dissonance

Plomp and Levelt (1965) conducted a series of experiments in which subjects were presented with pairs of sinusoidal tones at various frequency separations and were asked to rate the resulting sensation on a scale from consonant to dissonant. The principal empirical result was that perceived dissonance, for pairs of sinusoidal tones, depends not on the absolute frequency separation in hertz nor on the ratio of frequencies but on the separation as a fraction of the

critical bandwidth at the mean frequency of the pair. The dissonance function, normalized to unit amplitude at its maximum, is approximately:

$$D(\Delta f) = e^{-a\Delta f/CB} - e^{-b\Delta f/CB}$$

where Δf is the frequency separation, CB is the critical bandwidth at the mean frequency, and a and b are empirical constants approximately equal to 3.5 and 5.75 respectively. The function $D(\Delta f)$ is zero at $\Delta f = 0$ (no dissonance for identical tones), rises to a maximum at approximately $\Delta f = 0.25 \cdot CB$ (peak dissonance at one quarter of a critical bandwidth), and decreases monotonically thereafter, falling to negligible values for Δf greater than approximately one critical bandwidth.

The Plomp–Levelt dissonance function provides the empirical basis for the modern psychoacoustic theory of consonance. Its principal qualitative features are: that two sinusoidal tones are perceived as maximally dissonant when their frequency separation is approximately a quarter of the critical bandwidth at the mean frequency; that two sinusoidal tones are perceived as essentially consonant when their frequency separation is less than the critical bandwidth or substantially greater than the critical bandwidth; and that the dissonance is a monotonic function of frequency separation only within the critical bandwidth, beyond which it decreases. The function therefore identifies an interval of frequency separations within which roughness is perceived, and it locates the peak of that roughness at a particular fraction of the critical bandwidth.

8.3 Extension to Harmonic Tones: The Sethares Model

The Plomp–Levelt function characterizes the dissonance between pairs of sinusoidal tones. For pairs of harmonic tones—real musical tones, each consisting of a fundamental and an infinite series of upper partials—the total dissonance is, to a first approximation, the sum of the pairwise dissonances between every pair of partials, weighted by the product of their amplitudes. Sethares (1993, 1998) developed this approximation into a general computational framework for the prediction of dissonance from arbitrary timbres, and his framework has acquired the status of the standard psychoacoustic model of musical dissonance.

Formally, let tone A consist of partials at frequencies $f_A^{(1)}, f_A^{(2)}, \dots, f_A^{(N)}$ with amplitudes $a_A^{(1)}, a_A^{(2)}, \dots, a_A^{(N)}$, and similarly for tone B. The total Sethares dissonance is:

$$D_{\text{total}} = \sum_{i,j} a_A^{(i)} a_B^{(j)} D(|f_A^{(i)} - f_B^{(j)}|)$$

where the sum runs over all pairs of partials and D is the Plomp–Levelt dissonance function evaluated at the frequency separation between the partials. This formula reduces, in the limit where one tone has only its fundamental, to the Plomp–Levelt formula evaluated against the partials of the other tone; in the case where both tones have only fundamentals, it reduces to the Plomp–Levelt formula applied to the single pair. The Sethares formula therefore generalizes the Plomp–Levelt sinusoidal analysis to arbitrary spectral content.

Application of the Sethares formula to harmonic tones whose fundamentals stand in simple integer ratios recovers the conventional consonance ordering of the just-intonation intervals: the octave is most consonant, followed by the perfect fifth, the perfect fourth, the major sixth, and so on, in approximately the order given in the Pythagorean table. This recovery is not,

however, mathematically trivial: it is the consequence of the specific spectral structure of harmonic tones and the specific shape of the Plomp–Levelt dissonance function. For tones with non-harmonic spectra—inharmonic tones such as those produced by tubular bells, gamelan instruments, or stiff piano strings in the upper register—the Sethares formula predicts a different consonance ordering that does not correspond to simple integer ratios of fundamentals.

8.4 Inharmonic Timbres and Their Scales

The empirical claim that inharmonic timbres produce different consonance orderings has been substantiated by Sethares (1998) through analysis of the gamelan music of Java and Bali, in which the principal melodic instruments produce metallic tones with substantially inharmonic spectra and in which the scales used differ systematically from the just-intonation or equal-tempered scales of Western music. The pelog and slendro scales of Indonesian gamelan, with their distinctive intervals that do not correspond to simple-integer-ratio relationships in Western terms, may be understood, on the Sethares analysis, as the scales of minimum sensory dissonance for the specific inharmonic timbres of the instruments. The match between scale and timbre, in this sense, is not a cultural arbitrariness but a psychoacoustic optimization for the spectral content of the instruments in question.

A more striking analytical application of the Sethares framework is the design of artificial timbres for which any desired scale is the scale of minimum dissonance. Given an arbitrary scale of N notes per octave with arbitrary interval values, Sethares shows that a timbre may be constructed whose partials are positioned such that the scale notes correspond to the minima of the dissonance function. Music composed in the resulting matched scale–timbre system exhibits consonance properties analogous to those of conventional Western music in conventional Western timbres, but with intervals that are unrelated to small integer ratios. This analytical

possibility, demonstrated in compositions by Sethares and others, is the empirical disproof of any account of consonance that depends solely on the simple-integer-ratio structure of interval frequencies, independent of spectral content.

8.5 Harmonic Entropy and Related Models

An alternative quantitative model of consonance, due principally to Erlich (2000) and informally circulated under the title "harmonic entropy," attempts to characterize the consonance of an interval not by direct application of the Plomp–Levelt formula but by analysis of the entropy of the distribution of small-integer-ratio approximations to the given frequency ratio. The harmonic entropy of an interval is, roughly, the Shannon entropy of the distribution of nearby simple-integer ratios, weighted by their simplicity; consonant intervals are those whose harmonic entropy is low (because they are unambiguously identifiable with a single simple-integer ratio), and dissonant intervals are those whose harmonic entropy is high (because they admit many comparably accurate small-integer approximations).

The harmonic entropy model recovers the conventional consonance ordering with somewhat greater fidelity than the Sethares model in certain analytical respects, particularly for intervals near 600 cents (the tritone), and it provides an account of the categorical perception of musical intervals that the Sethares model does not. The two models are not, however, in direct competition: the Sethares model characterizes the sensory roughness of an interval as a function of its spectral content, while the harmonic-entropy model characterizes the cognitive identifiability of an interval as a function of its frequency ratio. The two are complementary aspects of consonance, and a complete account of perceived consonance presumably integrates both. The integration has not, at the time of writing, been carried out in a fully satisfactory mathematical form.

8.6 The Limits of Psychoacoustic Reduction

The psychoacoustic models surveyed in this chapter provide quantitative accounts of certain dimensions of consonance, but they do not, individually or collectively, exhaust the phenomenon. Three principal dimensions of consonance are not captured by the psychoacoustic models. First, the cognitive dimension: the perception of a musical interval is influenced by the listener's expectations, training, and prior exposure, none of which the sensory-roughness or harmonic-entropy models address. Second, the affective dimension: consonant intervals are not merely "smoother" than dissonant intervals but are perceived as carrying different emotional and aesthetic significance, which is not reducible to any of the psychoacoustic quantities. Third, the cultural dimension: the consonance ordering of intervals, while substantially universal across cultures for the most consonant and most dissonant cases, varies in its details across musical traditions, and the variations are not always predictable from the spectral content of the instruments in question.

These limitations do not impeach the psychoacoustic models as far as they go; they simply identify the boundaries of their analytical application. The simple-integer-ratio account of consonance, refined by the spectral analysis of Chapter VI and the psychoacoustic analysis of the present chapter, provides a substantially correct account of the sensory dimension of consonance for harmonic timbres in the Western tonal tradition. It does not provide a complete account of the phenomenon, and it does not pretend to. A complete account would integrate the sensory, cognitive, affective, and cultural dimensions into a unified theory; such a theory has not yet been developed in fully satisfactory mathematical form, and its development is one of the open problems of mathematical music theory.

8.7 Summary

This chapter has surveyed the psychoacoustic literature on consonance and dissonance. The principal results are: that the critical bandwidth of the basilar membrane characterizes the frequency-resolving precision of the auditory system and provides the perceptual scale on which consonance is measured; that the Plomp–Levelt dissonance function relates frequency separation to perceived dissonance and identifies a peak of dissonance at approximately one quarter of the critical bandwidth; that the Sethares model extends the Plomp–Levelt analysis to arbitrary timbres by summation over pairs of partials and recovers the conventional consonance ordering for harmonic tones; that inharmonic timbres produce different consonance orderings that do not correspond to simple-integer ratios of fundamentals; and that the psychoacoustic models, while substantively correct, do not exhaust the phenomenon of musical consonance, which has cognitive, affective, and cultural dimensions that the models do not capture. The bearing of these results on the principal analytical thesis of the present study is that the simple-integer-ratio account of consonance is properly understood as a corollary of the spectral and perceptual properties of harmonic tones, rather than as a freestanding mathematical diagnostic; the corollary status of the account does not impeach its substantial correctness for the music with which it is conventionally associated, but it does require that the account be situated within the broader spectral and perceptual framework rather than being treated as an independent mathematical principle.

Chapter IX

Synthesis and Critical Assessment

9.1 Restatement of the Analytical Position

The analytical investigation conducted in the preceding chapters supports the following composite position. The mathematics of musical consonance comprises four principal analytical instruments: the geometric sequence underlying twelve-tone equal temperament, the definite integral of the sinusoidal pressure waveform, the divergent series of integer-multiple harmonics, and the Fourier decomposition that relates time-domain signals to their spectral content. Each of these instruments contributes a partial and overlapping account of the phenomenon. None is sufficient on its own. None is, strictly speaking, dispensable. The composite account that emerges is one in which the perception of consonance arises from the interaction of three layers: a structural layer of integer-ratio frequency relations among harmonic spectra; a spectral layer of upper-partial coincidences and quasi-coincidences; and a perceptual layer of critical-bandwidth interactions and sensory roughness. Each layer is necessary; none is sufficient.

The simple-integer-ratio account, attributed to the Pythagorean tradition and codified in the just-intonation tabulation, is best understood as an articulation of the structural layer. It is substantially correct, for harmonic timbres in the Western tradition, that intervals whose fundamental frequencies stand in simple integer ratios are perceived as more consonant than intervals whose fundamental frequencies stand in complex ratios. But the simple-integer-ratio account does not explain why this is so. The explanation lies in the spectral layer: simple-integer-ratio intervals produce coincident upper partials in the harmonic spectra of the constituent tones, while complex-ratio intervals produce upper partials at close but non-coincident frequencies, and

these close-but-non-coincident frequencies produce sensory roughness as quantified by the Plomp–Levelt and Sethares analyses. The perceptual layer, in turn, mediates between the spectral structure and the conscious perception by means of the frequency-resolving properties of the cochlea and the central auditory pathway.

9.2 Critical Assessment of the IA Argument

The IA on which the present treatise is based developed three principal analytical claims: that the half-cycle integral provides a diagnostic of octave consonance; that the harmonic series of integer multiples corresponds to the perceptual notion of harmonic overtones; and that the simple-integer-ratio ordering of intervals reflects the density of overlapping overtones between the spectra of the constituent tones. The present work affirms the second and third of these claims, while requiring a substantial correction of the first.

The correction is as follows. The half-cycle integral of a unit-amplitude sinusoid is λ/π , where λ is the wavelength. The ratio of half-cycle integrals between two notes is therefore the ratio of their wavelengths, which by the elementary dispersion relation is the inverse of the ratio of their frequencies. The 2:1 ratio of half-cycle integrals between octave-related notes is, in this sense, an algebraic restatement of the 2:1 frequency ratio that defines the octave, rather than a freestanding diagnostic of octave consonance. The half-cycle integral does not provide additional information beyond what is contained in the frequency ratio itself, and it is therefore mistaken to characterize the 2:1 area ratio as a “cause” or “explanation” of octave consonance. The area ratio is a consequence, not a cause; an effect, not an explanation. The actual explanation, as developed in Chapters V, VI, and VIII, lies in the spectral coincidence of upper partials and in the corresponding minimization of sensory roughness.

This correction does not impeach the substantive insight of the IA argument; it refines the analytical claim that the argument was making. The substantive insight is that there is a deep mathematical structure underlying the perceptual phenomenon of consonance, and that the structure may be approached through elementary calculus. The substantive insight is correct. The analytical claim that the half-cycle integral provides a freestanding diagnostic is not. The proper analytical position, which the present inquiry adopts, is that the half-cycle integral provides a geometric visualization of the inverse-frequency relation, and that the inverse-frequency relation is, in turn, the structural precondition for the spectral coincidence that produces the perceptual consonance. The chain of explanation runs from spectral structure through perceptual processing to conscious perception, not from area ratio to consonance directly.

9.3 The Status of the Simple-Integer-Ratio Account

The simple-integer-ratio account of consonance, the principal analytical tradition examined throughout the foregoing chapters, admits the following final assessment. The account is substantially correct for harmonic timbres in the Western tonal tradition. It correctly orders the principal intervals by perceived consonance, with the octave most consonant, followed by the perfect fifth, the perfect fourth, the major and minor thirds and sixths, and so on, in approximately the order given by the increasing complexity of the corresponding simple-integer ratios. The account provides a heuristically powerful and pedagogically tractable framework within which the elementary structure of Western tonal music may be understood, and it constitutes the orthodox doctrine within the Western theoretical tradition for very good reasons.

The account is not, however, mathematically complete in the form in which it is usually presented. Its proper foundation, as developed in the present investigation, lies in the spectral coincidence of upper partials between harmonic tones, as analyzed by the Fourier-analytic

framework of Chapter VI and the psychoacoustic framework of Chapter VIII. The simple-integer-ratio account is a corollary of these more fundamental analyses; it is not an independent mathematical principle. The corollary status of the account has two principal consequences. First, the account applies only to harmonic timbres; for inharmonic timbres, the consonance ordering depends on the specific spectral content of the tones, and the corresponding scales of maximum consonance depart, often substantially, from the equal-tempered or just-intonation systems familiar from Western music. Second, the account is approximate even for harmonic timbres: the equal-tempered chromatic scale departs from just intonation by amounts that are spectrally and perceptually significant, and the consonance ordering of equal-tempered intervals is, in detail, somewhat different from that of just-intonation intervals.

These two consequences require that the simple-integer-ratio account be situated within a broader analytical framework rather than being treated as a freestanding doctrine. Within that broader framework, the account remains substantively correct and pedagogically valuable. It is not a complete theory of consonance, and it does not pretend to be; but it is a substantially correct first approximation, and the deviations from it admit principled analytical characterization through the more refined apparatus of Fourier analysis and psychoacoustics.

9.4 Open Questions

Three open questions arise from the analysis conducted herein and merit identification at the close of the synthesis. The first concerns the integration of the sensory, cognitive, and cultural dimensions of consonance into a unified mathematical theory. The sensory dimension is well characterized by the Plomp–Levelt and Sethares models; the cognitive dimension is less well characterized but admits partial treatment through the harmonic-entropy model of Erlich; the cultural dimension is studied principally in the ethnomusicological literature and has not yet been

formalized in a fully satisfactory mathematical framework. A unified theory that integrates all three would require analytical instruments substantially beyond those developed in the present work, and its development is one of the principal open problems of mathematical music theory.

The second concerns the analytical characterization of inharmonic timbres and their associated scales. The Sethares framework provides a constructive procedure for matching timbres to scales, but it does not, in its present form, provide a comprehensive theoretical account of which timbre-scale pairings are aesthetically successful. The empirical evidence from gamelan music and from the compositions in matched timbres developed by Sethares and others suggests that certain pairings are more aesthetically successful than others, but the analytical criteria by which the differential success may be predicted have not been fully articulated.

The third concerns the relationship between consonance perception and the broader cognitive processing of musical structure, including the perception of melody, the perception of harmony, and the perception of large-scale formal structure. The present investigation has restricted its attention to the pairwise consonance of simultaneously sounded tones, and the extension of the analytical framework to encompass the cognitive processing of more complex musical structures is a substantial open problem. The analytical instruments developed here—the Fourier-analytic framework, the psychoacoustic framework, and the elementary integral and series analyses of Chapters IV and V—provide some of the apparatus that such an extension would require, but the extension itself requires substantially more apparatus than is developed in the present pages, and its execution is properly the subject of a separate study.

9.5 Summary

This chapter has synthesized the analytical findings of the preceding chapters and has offered a critical assessment of the simple-integer-ratio account of consonance. The principal results are:

that the perception of consonance arises from the interaction of structural, spectral, and perceptual layers, each of which is necessary and none of which is sufficient; that the simple-integer-ratio account is best understood as an articulation of the structural layer, with the explanatory work performed by the spectral and perceptual layers; that the half-cycle integral analysis of the IA on which the present treatise is based requires correction in light of the inverse-frequency relation, which renders the area ratio a consequence rather than a cause of octave consonance; that the simple-integer-ratio account is substantively correct for harmonic timbres but is a corollary of the more fundamental spectral-coincidence and psychoacoustic accounts; and that the analytical framework developed here leaves open three principal questions concerning the integration of sensory, cognitive, and cultural dimensions; the characterization of inharmonic timbres; and the extension to the cognitive processing of complex musical structures.

Chapter X

Conclusion

10.1 Recapitulation

The present work has developed a mathematically rigorous account of the relationship between the calculus and the perception of musical consonance. The investigation has proceeded from the historical premises established in Chapter II through the mathematical apparatus developed in Chapters III through VI, the comparative analysis of tuning systems conducted in Chapter VII, and the psychoacoustic framework surveyed in Chapter VIII, to the critical synthesis offered in Chapter IX. The principal analytical instruments have been the geometric sequence underlying twelve-tone equal temperament, the definite integral of the sinusoidal pressure waveform, the divergent series of integer-multiple harmonics, and the Fourier decomposition that relates time-domain signals to their spectral content. Each instrument has been examined for its analytical reach and for its limitations; each has been situated within the broader historical and psychoacoustic literature; and each has been integrated into the composite account of consonance presented in Chapter IX.

10.2 Answers to the Research Questions

The five research questions identified in Section 1.5 admit the following answers in light of the analysis conducted.

On the first question—*to what extent does the simple-integer-ratio account of consonance admit principled derivation from the elementary apparatus of the calculus?*—the answer is that the simple-integer-ratio account is derivable from elementary calculus only as a

corollary of the spectral coincidence of upper partials between harmonic tones. The simple-integer-ratio ordering of intervals is a structural property of the harmonic series; the perceptual significance of that ordering is a consequence of the spectral and psychoacoustic considerations developed in Chapters VI and VIII. The elementary calculus apparatus, in itself, provides geometric visualization of the relations but not their perceptual explanation.

On the second question—*what is the mathematical relationship between the integral of the sinusoidal pressure waveform over one positive half-cycle and the perceived consonance of the corresponding musical tone?*—the answer is that the half-cycle integral of a unit-amplitude sinusoid is λ/π , where λ is the wavelength, and the ratio of half-cycle integrals between two notes is the ratio of their wavelengths and the inverse of the ratio of their frequencies. The half-cycle integral is therefore not a freestanding diagnostic of consonance but a geometric expression of the wavelength-frequency relation. It provides pedagogical visualization rather than analytical explanation.

On the third question—*in what sense does the divergent mathematical series of reciprocals correspond to the harmonic structure of an idealized vibrating string?*—the answer is that the mathematical harmonic series of reciprocals corresponds, term-for-term, with the sequence of wavelengths of the musical harmonic series of integer-multiple frequencies. The divergence of the mathematical series corresponds, in this analytical light, to the divergence of the partial-sum of overtone wavelengths or, equivalently, of half-cycle integrals. The perceptual significance, however, lies in the amplitudes of the overtones rather than in the frequencies or wavelengths as such, and the divergence of the mathematical series does not have a direct perceptual correlate.

On the fourth question—*how does Fourier analysis unify the heuristic observations of Chapters IV and V?*—the answer is that Fourier analysis recovers the simple-integer-ratio account as a consequence of spectral alignment between harmonic tones, while revealing the limitations of that account when applied to inharmonic timbres or refined by the psychoacoustic considerations of Chapter VIII. The Fourier-analytic framework is the formal apparatus within which the elementary observations of the earlier chapters acquire their proper analytical foundation, and it is the canonical framework within which subsequent analytical refinements are properly developed.

On the fifth question—*to what extent does the simple-integer-ratio account remain defensible when measured against the psychoacoustic literature?*—the answer is that the simple-integer-ratio account is substantively correct for harmonic timbres but is properly understood as a corollary of the spectral and perceptual properties of harmonic tones, rather than as a freestanding mathematical diagnostic. The account requires generalization to accommodate inharmonic timbres and refinement to accommodate the perceptual phenomena of critical bandwidth and sensory roughness, but its principal qualitative conclusions are recovered by the more refined apparatus and the principal historical insight is therefore vindicated, in a clarified analytical form, by the modern psychoacoustic literature.

10.3 Directions for Further Research

The analytical investigation conducted in the preceding chapters identifies several directions for further research that warrant brief mention at the close.

First, the extension of the Fourier-analytic framework from the periodic case treated here to the time-evolving signals characteristic of real musical performance requires the apparatus of short-time Fourier analysis, wavelet analysis, and time-frequency distributions more generally.

These analytical instruments have been developed extensively in the digital signal processing literature but have not been systematically integrated with the theory of consonance as presented in the present pages. A separate investigation is warranted.

Second, the integration of the sensory dimension of consonance, characterized by the Plomp–Levelt and Sethares models, with the cognitive dimension characterized by the harmonic-entropy model and with the cultural dimension characterized by the ethnomusicological literature, remains an open problem. A unified theory that integrates the three dimensions would require analytical instruments substantially beyond those developed here, including potentially the apparatus of probabilistic generative modeling, statistical learning theory, and computational neuroscience.

Third, the extension of the analytical framework from pairwise consonance to the consonance of larger note-collections—chords, harmonic progressions, and full musical phrases—requires the systematic treatment of multi-dimensional spectral interactions, of voice-leading constraints, and of the temporal cognitive processing of musical structure. This extension is the proper subject of a music-theoretic rather than a strictly mathematical treatise, but the mathematical apparatus developed here provides some of the analytical instruments such a study would require.

Fourth, the systematic application of the Sethares framework to the design of new musical timbres and to the corresponding development of new scales and tuning systems presents itself as an open compositional and analytical program. The technological apparatus required—digital synthesis of arbitrary spectral content, real-time tuning adjustment, computer-assisted composition—is now widely available, but the aesthetic and theoretical principles by

means of which the analytical possibilities of the framework may be converted into compelling musical practice have not been comprehensively articulated.

10.4 Closing Observations

The proposition with which this inquiry began—that the calculus contributes substantively to the mathematical characterization of musical consonance—admits the affirmative answer that the investigation has substantiated. The calculus provides the elementary apparatus through which the geometric sequence of equal temperament, the integral of the sinusoidal waveform, and the series of harmonic overtones are formally characterized. The calculus does not, however, provide the complete framework within which consonance is to be understood: that framework requires the Fourier-analytic apparatus of the nineteenth century and the psychoacoustic apparatus of the twentieth, together with the comparative analysis of historical tuning systems and the critical assessment of the simple-integer-ratio tradition. The calculus is, in this sense, necessary but not sufficient for the analytical investigation of musical consonance.

The investigation conducted in these chapters has aimed to articulate the contribution of the calculus with precision, to integrate that contribution with the broader analytical framework, and to identify with equal precision the limits of what the calculus can deliver. The investigation has not, and could not, exhaust the subject. The mathematical theory of musical consonance is one of the oldest open problems of Western thought, and it remains, in the third decade of the twenty-first century, open in its detailed formulations and in its integration of sensory, cognitive, and cultural dimensions. The present work represents one contribution to that long-standing investigation. It does not claim more, and it does not claim less.

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Appendix A

Derivations and Proofs

A.1 Derivation of the Equal-Tempered Semitone Ratio

Let r denote the equal-tempered semitone ratio. The closure condition of the octave requires that twelve successive applications of r return a tone of exactly double the fundamental frequency:

$$r^{12} = 2$$

Taking the twelfth root of both sides:

$$r = 2^{1/12}$$

The value $2^{1/12}$ is irrational; its decimal expansion begins 1.0594630943592953....

The irrationality may be established by the elementary observation that if $r = p/q$ were rational with p and q coprime positive integers, then $p^{12} = 2 q^{12}$, which would require p to be even and hence q to be even, contradicting coprimality. The cents-value of one equal-tempered semitone is therefore $1200 \cdot \log_2(2^{1/12}) = 100$ cents, exactly, by the definition of the cent.

A.2 Half-Cycle Integral of the Sinusoid

Let $y(x) = \sin(2\pi x / \lambda)$ be the unit-amplitude sinusoid with wavelength λ . The integral over one positive half-cycle, from $x = 0$ to $x = \lambda/2$, may be computed by elementary substitution. Let $u =$

$2\pi x / \lambda$. Then $du = (2\pi / \lambda) dx$, so $dx = (\lambda / 2\pi) du$. When $x = 0$, $u = 0$; when $x = \lambda/2$, $u = \pi$. The integral therefore reduces to:

$$\int_0^{\lambda/2} \sin(2\pi x / \lambda) dx = \frac{\lambda}{2\pi} [-\cos(2\pi x / \lambda)]_0^{\lambda/2} = \frac{\lambda}{2\pi} \cdot 2 = \frac{\lambda}{\pi}$$

The half-cycle integral of the unit-amplitude sinusoid is therefore exactly λ / π . The corresponding integral for a sinusoid of amplitude A is $A \cdot \lambda / \pi$, by linearity of the integral.

A.3 Closed-Form Cents-Value of the Pythagorean Comma

The Pythagorean comma is the ratio of twelve perfect fifths to seven octaves:

$$PC = \frac{(3/2)^{12}}{2^7} = \frac{3^{12}}{2^{12} \cdot 2^7} = \frac{531441}{524288}$$

The cents-value is:

$$PC_{\text{cents}} = 1200 \cdot \log_2 (531441/524288) \approx 1200 \cdot \log_2 (1.013643) \approx 23.460 \text{ cents}$$

A.4 Derivation of Fletcher's Inharmonicity Formula for Stiff Strings

For a real piano string of nonzero stiffness, the n -th mode frequency departs from the ideal integer-multiple value by a small correction that grows with mode number. Fletcher (1964) derived the approximate formula:

$$f_n = n \cdot f_1 \cdot \sqrt{1 + B n^2}$$

where B is the inharmonicity coefficient, given in terms of the string parameters by $B = \pi^3 E d^4 / (64 T L^2)$, with E denoting the Young's modulus, d the string diameter, T the tension, and L the speaking length. For typical piano strings, B ranges from approximately 10^{-4} for the lowest bass strings to approximately 10^{-2} for the highest treble strings. The inharmonicity correction is the principal reason for the stretched tuning of the piano, in which the upper register is tuned slightly sharp and the lower register slightly flat relative to equal temperament.

Appendix B

Frequency Tables for Equal Temperament

The following table provides equal-tempered frequencies for the chromatic scale across nine octaves, from C0 (16.35 Hz) to B8 (7902.13 Hz), computed at A4 = 440 Hz with $r = 2^{(1/12)}$.

Frequencies are rounded to two decimal places.

Table B1. Equal-Tempered Frequencies (Hz) for Octaves 0–8

Note	0	1	2	3	4	5	6	7	8
C	16.35	32.70	65.41	130.81	261.63	523.25	1046.50	2093.00	4186.01
C#	17.32	34.65	69.30	138.59	277.18	554.37	1108.73	2217.46	4434.92
D	18.35	36.71	73.42	146.83	293.66	587.33	1174.66	2349.32	4698.64
D#	19.45	38.89	77.78	155.56	311.13	622.25	1244.51	2489.02	4978.03
E	20.60	41.20	82.41	164.81	329.63	659.26	1318.51	2637.02	5274.04
F	21.83	43.65	87.31	174.61	349.23	698.46	1396.91	2793.83	5587.65
F#	23.12	46.25	92.50	185.00	369.99	739.99	1479.98	2959.96	5919.91
G	24.50	49.00	98.00	196.00	392.00	783.99	1567.98	3135.96	6271.93
G#	25.96	51.91	103.83	207.65	415.30	830.61	1661.22	3322.44	6644.88
A	27.50	55.00	110.00	220.00	440.00	880.00	1760.00	3520.00	7040.00
A#	29.14	58.27	116.54	233.08	466.16	932.33	1864.66	3729.31	7458.62
B	30.87	61.74	123.47	246.94	493.88	987.77	1975.53	3951.07	7902.13

Appendix C

Glossary of Technical Terms

The following glossary defines the principal technical terms used throughout the foregoing chapters. Definitions are provided in the analytical sense intended in the text and may differ in detail from definitions encountered in other technical literatures.

Beat. The slow periodic variation in amplitude produced by the superposition of two sinusoids of slightly different frequencies. The beat frequency is the difference of the two component frequencies, and the beat envelope is itself sinusoidal.

Cent. One one-hundredth of an equal-tempered semitone; the standard logarithmic unit of measurement for small interval differences. One octave equals 1200 cents by definition.

Consonance. The perceived quality of two or more simultaneously sounded musical tones in which the combined sonority is felt as stable, restful, or pleasing. The structural correlate is simple-integer ratio of fundamental frequencies; the spectral correlate is coincidence of upper partials; the perceptual correlate is the absence of sensory roughness.

Critical bandwidth. The bandwidth of an auditory filter at a given frequency; the bandwidth within which two pure tones interact perceptually to produce roughness or masking. Approximately constant at low frequencies and proportional to frequency at high frequencies.

Dissonance. The perceptual complement of consonance: the perceived quality of simultaneously sounded tones in which the combined sonority is felt as unstable, tense, or unpleasant. The

principal sensory correlate is roughness arising from the interaction of upper partials within the critical bandwidth.

Equal temperament. The tuning system in which the octave is divided into twelve equal logarithmic intervals, producing a chromatic scale with the semitone ratio $r = 2^{(1/12)}$. The standard tuning system of modern Western music.

Fourier series. The representation of a periodic function as an infinite trigonometric series whose terms are sinusoids of frequencies that are integer multiples of the fundamental frequency.

Fundamental. The lowest frequency component of a harmonic tone; the frequency that determines the perceived pitch. Distinguished from the upper partials or overtones, which determine the timbre.

Harmonic series (mathematical). The divergent infinite series $\sum 1/n$. Distinguished from the musical harmonic series.

Harmonic series (musical). The indexed sequence of integer multiples of a fundamental frequency, corresponding to the standing-wave modes of an idealized vibrating string with fixed endpoints. The basis of the spectral content of a harmonic tone.

Inharmonicity. The property of a vibrating system whose normal-mode frequencies are not exactly integer multiples of the fundamental, typically as a consequence of nonzero stiffness in the vibrating medium.

Just intonation. A tuning system in which the principal intervals are realized as exact simple-integer ratios of frequency. In its standard 5-limit form, the relevant primes are 2, 3, and 5.

Octave. The musical interval between two tones whose fundamental frequencies stand in the ratio 2:1. The most consonant interval other than the unison.

Overtone. Any frequency component of a tone above the fundamental. The n -th overtone is the $(n+1)$ -th harmonic in the musical harmonic series.

Pythagorean comma. The closure error of the cycle of perfect fifths in Pythagorean tuning, equal to $(3/2)^{12} / 2^7 \approx 1.01364$, or approximately 23.46 cents.

Syntonic comma. The discrepancy between four perfect fifths and a major third plus two octaves, equal to $81/80 \approx 1.0125$, or approximately 21.51 cents. The principal complication in 5-limit just intonation.

Timbre. The perceived quality of a tone that distinguishes it from another tone of the same pitch and loudness; determined principally by the amplitude distribution of the harmonic spectrum.

Wavelength. The spatial period of a sinusoidal wave; the distance between successive maxima of the waveform. Related to frequency and velocity by the dispersion relation $\lambda = v/f$.